FACULTY OF ECONOMICS AND BUSINESS ADMINISTRATION

Multinational networks and natural disasters

Nathan Szelagowski
Student number: 02401732

Supervisor: Prof. dr. Bruno Merlevede

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Nathan Szelagowski, August 2026

FOREWORD

This dissertation closes a two-year master's programme at Ghent University, and with it five years as a business engineering student. The work owes a great deal to the people who accompanied it, and it seems right to name them before the reader turns to the substance.

My first thanks go to my promoter, Professor Bruno Merlevede. He gave generously of his time throughout, and he made himself available whenever a question needed settling. Each meeting left the project clearer than it had been an hour earlier, and the direction he offered at those moments is what allowed the work to keep moving rather than stall. This thesis would not have taken the shape it has without him.

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To my friends and my family, thank you for the support that had nothing to do with econometrics and everything to do with getting to the end of the year. It mattered more than you probably realise.

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Nathan Szelagowski
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LIST OF USED ABBREVIATIONS

Abbreviation	Definition
AME	Average marginal effect
CEPII	Centre d'Études Prospectives et d'Informations Internationales
CRED	Centre for Research on the Epidemiology of Disasters
EM-DAT	Emergency Events Database
EMP	Number of employees (FATS variable)
ENT	Number of enterprises, that is, foreign affiliates (FATS variable)
FATS	Foreign Affiliates Statistics
FDI	Foreign direct investment
GDP	Gross domestic product
GE	Government effectiveness
GVC	Global value chain
HC3	Heteroskedasticity-consistent covariance matrix estimator, type 3
IRLS	Iteratively reweighted least squares
ISIC	International Standard Industrial Classification, Rev. 4
LP	Local Projections
LPM	Linear probability model
MNE	Multinational enterprise
NACE	Statistical classification of economic activities in the European Community, Rev. 2
OECD	Organisation for Economic Co-operation and Development
OLI	Ownership, Location, Internalisation paradigm
OLS	Ordinary least squares
PPML	Poisson pseudo-maximum likelihood
RQ	Research question
SE	Standard error
TiVA	Trade in Value Added
TUR	Turnover (FATS variable)
TWFE	Two-way fixed effects
UCLouvain	Université catholique de Louvain
USD	United States dollar
VAR	Vector autoregression
VIF	Variance inflation factor
WDI	World Development Indicators
WGI	Worldwide Governance Indicators
WHO	World Health Organization

ABSTRACT

This dissertation asks how the presence of European multinational affiliates relates to natural disaster exposure in host countries, and whether institutional quality and value chain position condition that relationship.

It builds a dyadic panel of European outward foreign affiliate statistics covering 2008 to 2019, matched to disaster records from EM-DAT, governance and macroeconomic indicators from the World Bank, value chain position from OECD Trade in Value Added, and bilateral geography from CEPII. Three questions follow. The first asks whether accumulated disaster exposure is associated with where affiliates are located. The second asks whether affiliate presence recovers in the years following a sudden shock. The third asks whether a sector's position in the global value chain moderates that response. Estimation proceeds by cross-sectional gravity models for the first question and by Poisson pseudo-maximum-likelihood local projections for the other two.

Host countries with a longer record of reported disasters show a greater presence of European affiliates although institutional quality attenuates it. No post-shock rebound appears at any horizon within five years. The value chain question returns no reliable answer: one significant coefficient emerges among more than a hundred estimates and is reproduced under neither an alternative estimator nor an alternative outcome measure.

These results are bounded less by estimation than by measurement. Reported disaster frequency in EM-DAT covaries with the same governance characteristics that this literature uses as moderators, so a cross-section of this kind cannot separate physical hazard from reporting capacity. That constraint, documented here for this sample, applies to a wider body of cross-country disaster research.

1 INTRODUCTION

1.1 Problem Statement and Research Objective

Multinational enterprises (MNEs) hold physical and organisational assets (factories, warehouses, supplier relationships) spread across dozens of jurisdictions, following decades of fragmentation of production into globally dispersed value chains (Antràs & Chor, 2022). A substantial share of those locations is exposed to weather-related hazards such as floods, storms and wildfires, and a further share to geophysical hazards such as earthquakes and volcanic activity, whose incidence is unrelated to climate but no less consequential for the firms operating there. How multinationals respond when a host country is struck by such a shock therefore speaks directly to the resilience of the cross-border production networks on which a large part of the world economy now depends.

Existing research on this question is organised at three levels of observation. A first, macroeconomic strand relates aggregate FDI inflows to country- or sector-level disaster measures, and has not converged on a sign: Escaleras & Register (2011) report a reduction in inflows following disasters, whereas Neise et al., (2022) find sector-dependent effects and a positive association between disaster exposure and FDI inflows three to five years after an event, and Doytch (2020) finds negative effects overall that vary with disaster type, industry and region. A second strand works at the level of the affiliate. Using data on the foreign affiliates of 71 European MNEs, Oh & Oetzel (2011) find that major natural disasters have no significant effect on the number of affiliates a firm maintains, in contrast to terrorist attacks and technological disasters which do reduce it, while host-country regulatory quality shapes the investment response specifically to natural events.Bijnens et al. (2024) track individual Belgian firms around the 2021 floods and document persistent sales losses concentrated among firms whose suppliers were affected. A third strand identifies the propagation of disaster shocks through production networks: Barrot & Sauvagnat (2016) show that natural disasters striking suppliers impose substantial output losses on their customers, particularly where inputs are specific, and Boehm et al. (2019) document sharp production declines in the US affiliates of Japanese multinationals after the 2011 Tōhoku earthquake.

Two gaps motivate this thesis. First, none of these literatures varies hazard type, host-country institutional environment and value-chain position within a single empirical framework, and the null result of Oh & Oetzel (2011) has not been re-examined across a substantially broader set of host countries and industries. Second, cross-country work in this area measures exposure almost exclusively through events recorded in EM-DAT, whose reporting completeness varies systematically across countries (Delforge et al., 2025), and varies with the same country characteristics that these studies use as moderators. This thesis documents that variation for its own sample (see section 3.1.1).

Conceptually, the thesis departs from the assumption that the physical magnitude of a shock is the main determinant of firm response and follows the view that a host economy's accumulated capacity to anticipate and absorb shocks is at least as decisive. The strongest support for this view is provided by Hsiang & Jina (2014), who find that cyclone-induced losses are amplified in countries with less historical exposure to such events. The first research question asks whether that accumulated capacity is associated with where affiliates are located at all. The second and third research questions are whether accumulated exposure provides the same protection to MNE affiliates. The three research questions are organized along two margins: the ex-ante location decision which a firm can only evaluate through a host country’s externally observable institutions, i.e. the location component of Dunning’s OLI paradigm (Dunning & Lundan, 2008) and post-entry adjustment where the relevant capacity is the firm’s own accumulated experience, closer to internalization theory in its more recent form (Verbeke & Kano, 2015). This is a distinction made for the purpose of organizing the three questions, not an object of analysis itself.

The analysis draws on a dyadic panel merging Eurostat's Foreign Affiliates Statistics (FATS) with the EM-DAT disaster database maintained by CRED, World Bank governance and macroeconomic indicators, OECD Trade in Value Added data, and CEPII gravity variables, covering 29 European home countries and up to 185 host countries over 2008–2019, for 104,453 dyad-year-industry cells. The data are aggregated: individual firms are not observed, only affiliate and employee counts within each home–host–industry cell. Three research questions are examined. RQ1, on the ex-ante location margin, is addressed on a pair-level cross-section. RQ2 and RQ3, on post-shock adjustment, are addressed by Local Projections (Jordà, 2005) over horizons of five and three years respectively, using both a log-linear specification capturing adjustment among active affiliates and a Poisson pseudo-maximum-likelihood specification (Santos Silva & Tenreyro, 2006) that retains cells falling to zero. The latter approximates market exit at the cell level without identifying the exit of any given affiliate. The answers run against the account that dominates each of the three literatures, and the pattern is itself informative. On RQ1, historical disaster exposure is associated with more affiliate activity, not less, and government effectiveness dampens rather than amplifies this positive association. On RQ2, no creative-destruction rebound appears at any horizon in either specification, and the estimated trajectory runs opposite to the one that account implies. RQ3 yields no robust estimates, its triple-interaction design being constrained by the 56 host countries for which value-chain data are available. The thesis therefore makes three contributions. It provides a broad-coverage test that corroborates Oh & Oetzel (2011)’s null result for natural disasters while failing to replicate, at the affiliate level, the delayed FDI rebound reported by Neise et al. (2022). It documents how the measurement properties of EM-DAT exposure data covary with the institutional variables this literature uses as moderators, and offers a measurement reading of these estimates. And it brings exposure history, institutional quality and value-chain position into a common framework, establishing what such a framework can and cannot identify with currently available aggregated data.

1.2 Thesis Structure

The remainder of the thesis is structured as follows. Chapter 2 reviews the theoretical and empirical literature on MNE affiliate behaviour, natural disasters, and global value chains, and develops the three research questions introduced above. Chapter 3 details the data sources, sample construction, and econometric strategy. Chapter 4 presents the empirical results and discusses them critically in light of the data's limitations. Chapter 5 concludes with the thesis's contribution, its policy implications, and directions for further research.

2 LITERATURE REVIEW

2.1 Overview

The question this thesis addresses rests on a premise worth stating at the outset: what governs how a multinational affiliate responds to a natural disaster is not, in the first instance, the physical magnitude of the shock, but the accumulated capacity of the location to anticipate and absorb it. A hazard of given intensity does not translate mechanically into a given loss. It is mediated by prior investment in prevention, by the institutional environment that governs reconstruction, and by the position the affiliate occupies within the production network to which it belongs. This chapter develops that premise in four steps. Section 2.2 sets out the theory of the multinational firm and the distinction between the entry margin and post-entry adjustment that organises what follows. Sections 2.3 to 2.5 then review three empirical strands (location under institutional risk, post-shock dynamics, and propagation through value chains) each formulating one of the thesis's three research questions. Section 2.6 states the gaps these literatures leave open and summarises what they do and do not settle.

2.2 The theory of the multinational firm

The location component of the OLI paradigm holds that a firm establishes production abroad when a host location offers an advantage it cannot obtain elsewhere (Dunning & Lundan, 2008). What matters for the present purpose is the informational asymmetry embedded in that decision. Before entry, a firm has no operating experience of the location; what it can assess is confined to what is externally observable, and the formal institutional environment (the quality of public administration, the enforceability of contracts, the credibility of regulation) is the principal such signal available to it. The entry margin is therefore an institutional margin.

Post-entry adjustment is governed by a different logic. Once established, a firm accumulates location-specific knowledge and reorganises its affiliate structure in response to shocks under bounded rationality and adaptive governance, which is the domain of internalisation theory in its more recent form (Verbeke & Kano, 2015). This distinction organises the three research questions below. It is a framing device rather than an object of analysis: the data used in this thesis cannot arbitrate between the two frameworks, and no such arbitration is claimed.

Two further points fix the unit of analysis. First, foreign affiliates do not behave as local firms do under shock, as they have access to the parent’s resources a local firm wouldn’t have, which is what makes the affiliate rather than the host economy the relevant object of study (Alfaro & Chen, 2012). Second, adjustment can occur along two margins: an intensive margin, through the scale of activity within continuing affiliates, and an extensive margin, through the entry and exit of affiliates themselves. The decomposition is native to work on multinational networks: Merlevede & Michel (2024) treat a change in the number of affiliates in a network as movement at its extensive margin, and a change in the activity of the affiliates that remain as movement at its intensive margin. This thesis observes both, and Chapter 3 sets out the estimators that separate them.

2.3 Disasters, institutions and location

2.3.1 Why firms locate in exposed places

That firms operate in hazard-prone locations at all requires explanation before any penalty can be posited. Dunning's motives (resource-seeking, market-seeking, efficiency-seeking) imply that a firm will accept a known risk where the location offers an advantage unavailable elsewhere (Dunning & Lundan, 2008). Economic geography sharpens the point considerably. Exposed locations are

disproportionately coastal, deltaic or seismically active; that is to say, they are ports, river mouths and dense agglomerations. The same physical features that generate hazard exposure (proximity to water, to fault-adjacent mountain corridors, to the alluvial plains on which large cities were historically founded) are the features that generate agglomeration economies, market access and transport advantage. The location advantage and the hazard are not independent draws; they are frequently two properties of the same site. A positive association between historical exposure and multinational presence is therefore not, on its face, anomalous.

2.3.2 The macroeconomic literature has not converged on a sign

Aggregate work relating FDI to country- or sector-level disaster measures has produced no consensus. Escaleras & Register (2011) report a reduction in FDI inflows following disasters. Doytch (2020), examining sectoral inflows across 69 countries between 1980 and 2011, finds negative effects overall but effects that vary with disaster type, industry and region. Neise et al. (2022), using hybrid panel regressions across 181 countries and four economic sectors, find that the effect varies by sector, and that a negative short-run response is followed by an increase in inflows three to five years after an event. The sign is thus contested, and it is contested in a way that appears to depend on the level of aggregation, the sector, and the horizon considered.

One feature these studies share deserves note. Exposure is measured throughout by counts of reported events drawn from the same source, and the reporting properties of that source are not neutral with respect to the country characteristics this literature uses as moderators. Section 3.1.1 examines those properties for the present sample.

2.3.3 A theoretical account of multinational resilience

Kato & Okubo (2022) provide the most developed theoretical treatment of the question. Their contribution is a model, not an empirical exercise: foreign multinationals and local firms in a host country are linked through input–output relations, and a natural disaster is modelled as an increase in the fixed cost borne by local suppliers. Because multinational profitability depends on the price of locally sourced intermediates, and that price depends in turn on the mass of surviving local suppliers, the system admits multiple equilibria: one in which multinationals and local suppliers coexist, one in which multinationals are absent. A sufficiently large shock reduces the supplier base below the threshold at which multinational operation remains profitable, and the economy switches from the first configuration to the second; under myopic location decisions, the switch is permanent. The model's central result concerns the conditions under which the switch does not occur: multinationals are more likely to remain where local intermediates account for a larger share of their production costs, and where the trade costs of importing foreign inputs are lower.

The paper's empirical section precedes the model and is presented by the authors as suggestive motivation rather than as evidence in its own right. Regressing country-level FDI inflows on counts of EM-DAT events over 1991–2015, they obtain positive and significant coefficients on the disaster variables, which they read as capturing either the return of temporarily displaced investment or the opportunity that reconstruction affords for capital upgrading. The interaction between the disaster count and a developing-country indicator is negative and significant.

A further extension bears directly on the location margin. Allowing forward-looking firms to face uncertainty over the timing of future events, Kato and Okubo show that disaster risk alone can induce multinationals to leave a host country in anticipation, and that the expected duration of presence declines as the perceived frequency of events rises. Exposure history, on this account, operates as a signal about the future rather than only as a record of the past.

2.3.4 Affiliate-level evidence

The closest antecedent at the level of the affiliate is Oh & Oetzel (2011), who use data on the foreign affiliates of 71 European multinationals. They find that major natural disasters have no significant effect on the number of affiliates a firm maintains, in contrast to terrorist attacks and technological disasters, which do reduce it. Host-country regulatory quality shapes the *investment* response specifically to natural events. Their null result on affiliate counts has not, to our knowledge, been re-examined across a substantially broader set of host countries and industries.

2.3.5 From the literature to RQ1

Two considerations pull in opposite directions. The geography of exposure, set out above, implies that hazard-prone locations frequently carry offsetting location advantages, so a positive association between exposure and affiliate presence cannot be ruled out ex ante. Against this, the risk extension in Kato & Okubo (2022) supplies a theoretical basis for expecting a penalty: where the perceived frequency of events is higher, the expected duration of multinational presence is shorter, and this holds independently of any realised shock. The literature therefore does not settle the sign of the association, and the two mechanisms it identifies point in opposite directions. Which dominates is an empirical question rather than one the theory answers, and it is the first this thesis asks. The role of institutional quality follows from the informational structure of the entry decision. A firm contemplating entry cannot observe how a location will in fact perform under stress; it can observe the institutions that will govern warning, response, reconstruction and compensation. Strong governance may therefore substitute for the experience the firm lacks, offsetting an unfavourable exposure history. One measurement point qualifies what follows. The location margin is observed here as the cross-sectional distribution of affiliate presence over the panel window, which reflects cumulative entry *and* survival rather than entry alone.

RQ1: To what extent is a host country's historical exposure to natural disasters associated with the level of European MNE affiliate presence, and does host-country government effectiveness moderate that association?

2.4 Post-shock dynamics and creative destruction

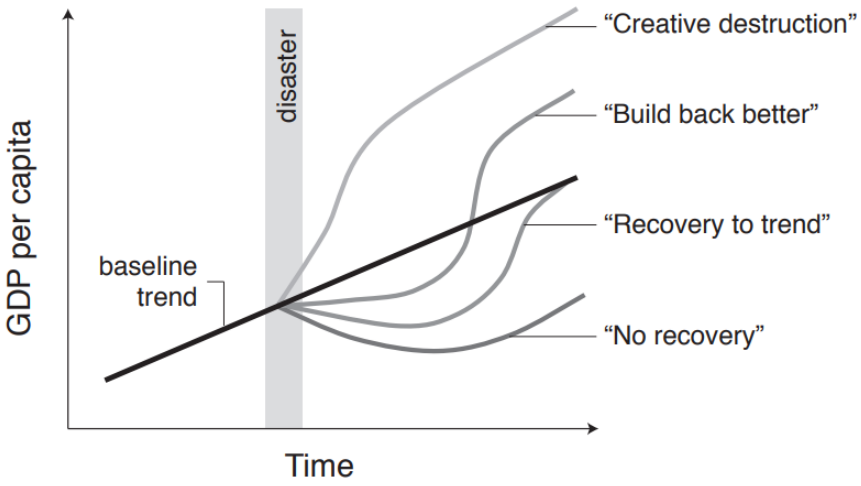


Figure 2-1: "Four hypotheses, proposed in the literature, that describe the long-term evolution of GDPpc following a natural disaster" (Hsiang & Jina, 2014, Figure 1).

2.4.1 The creative-destruction proposition

A long-standing proposition holds that disasters can raise long-run output by forcing the replacement of obsolete capital. Skidmore & Toya (2002) argue that the destruction of physical capital accelerates its turnover, shifting the capital stock toward more recent and more productive vintages and raising the growth path. The proposition has received qualified empirical support at best. Crespo Cuaresma et al. (2008). find the mechanism operating only among the wealthier developing countries in their sample and absent among the poorest, a result that constrains the proposition rather than confirming it: capital upgrading appears to require a level of income and absorptive capacity that the countries most exposed to catastrophic loss frequently lack.

2.4.2 The rejection in the recent growth literature

More recent work is largely unfavourable to the proposition. Felbermayr & Gröschl (2014), constructing measures of physical disaster intensity in order to avoid the endogeneity of reported damages to income and institutions, find robustly negative growth effects. Hsiang & Jina (2014) trace national income following tropical cyclones and find that affected economies decline relative to their pre-disaster trajectory and do not recover over a twenty-year horizon. On this evidence, the creative-destruction proposition is contested at the level of aggregate output.

2.4.3 From aggregate output to the FDI margin

The four paths set out in Figure 2-1 share a feature on which the step to the present question turns: each is defined over aggregate output per capita. What separates them — whether the post-disaster path overshoots the counterfactual trend, returns to it, or settles permanently below it — is a disagreement about that single quantity, and the evidence reviewed above adjudicates it in favour of the lowest path. The margin examined in this thesis is not that quantity. Foreign investment can recover where domestic output does not: reconstruction generates demand, modernised capital raises the return to new entry, and inward investment may rise while the national income path remains depressed. A finding of no recovery in GDP per capita and a rebound in multinational presence are therefore not in contradiction. They concern different objects, and settling the first does not settle the second.

Kato & Okubo (2022) supply the microfoundation for such a rebound. Extending their model to a temporary rather than permanent shock and to forward-looking location decisions, they characterise the timing of re-entry: multinationals return to a recovering host country the sooner the faster reconstruction proceeds, the greater their dependence on local intermediates, and the lower the trade costs they face on imported inputs. Where recovery is slow enough, re-entry does not occur at all. Empirically, the closest antecedent is Neise et al. (2022), whose finding of an increase in FDI inflows three to five years after an event is precisely what RQ2 examines, at a finer level of observation and over the same horizons. One caveat on the comparison is important. Neise et al. measure inflows, which are a flow; this thesis measures affiliate and employment counts, which are a stock. A rebound in inflows is compatible with a flat affiliate count, since inflows include reinvestment in affiliates that already exist. The two measures answer related but distinct questions.

2.4.4 From the literature to RQ2

If a rebound occurs, its magnitude should depend on how much obsolete capital the shock destroyed and how effectively the host economy can rebuild, both of which are conditioned by prior exposure. Hsiang & Jina (2014) provide the relevant evidence: cyclone-induced losses are amplified in countries with less historical exposure to such events, which is consistent with locations that have experienced repeated events having already built to a standard that limits marginal destruction. Where initial

destruction is muted, there is correspondingly less obsolete capital to replace, and the modernisation channel should be weaker.
What the empirical design examines is the reduced-form pattern implied by this reasoning (a rebound differing systematically with historical exposure) and not the capital-vintage mechanism itself, which is not observable in counts of affiliates and employees.

RQ2: How does European MNE affiliate presence in a host country evolve over the five years following a sudden natural disaster, and does that trajectory differ with the host country's historical exposure to such events?

2.5 Propagation and vulnerability in global value chains

2.5.1 Fragmentation and position in the chain

Production has been fragmented into stages distributed across countries, and the governance of each stage is shaped by the incompleteness of the contracts that link them (Antràs & Chor, 2022). Position within the resulting sequence is not a neutral attribute. Upstreamness — the distance of an activity from final demand — determines how many downstream stages depend on a given node's output, and how readily the inputs that node supplies can be substituted. Position is therefore a determinant of exposure to disruption risk, independently of the scale of the activity concerned.

2.5.2 The evidence on propagation

That disruption travels along these links is well established. Barrot & Sauvagnat (2016) show that natural disasters striking suppliers impose substantial output losses on their customers, with losses concentrated where the inputs concerned are specific and therefore difficult to replace at short notice. Boehm et al. (2019) document sharp production declines in the US affiliates of Japanese multinationals following the 2011 Tōhoku earthquake, and Carvalho et al. (2021) trace comparable propagation through the Japanese domestic network after the same event.
The relevant conclusion for what follows is not that upstream nodes are fragile, but that disruption at an upstream node is costly well beyond the site affected. The loss falls on the network, not only on the establishment struck.
One scope condition must be stated. These studies identify the transmission of a shock occurring in one location to firms in others. RQ3 asks a different question: whether an affiliate's own position in the chain moderates the effect of a disaster occurring in its own location. The exposure channel is distinct, and the propagation evidence establishes the network cost of an upstream disruption rather than the location-specific vulnerability of an upstream affiliate.
One study addresses that channel directly. Altay & Ramirez (2010) analyse more than 3,500 disasters against over 100,000 firm-year observations and find that the effect of flooding on a firm's total asset turnover depends on its position in the supply chain, upstream firms experiencing a positive impact and downstream firms the opposite. The result is established at firm level within a broad cross-section of sectors rather than across host countries, and it concerns operating performance rather than continued presence.

2.5.3 From the cost of propagation to retention

It is precisely because upstream disruption is costly at network scale that the multinational's incentive runs toward repair rather than exit. Upstream affiliates typically embody higher asset specificity and larger sunk investments, and the option value of retaining a node whose loss would propagate through the network is correspondingly high. Downstream affiliates, by contrast, are more footloose:

relocation remains comparatively cheap, and the systemic consequence of their closure is contained. The relevant contrast is therefore one of differential retention rather than differential vulnerability. Kato & Okubo (2022) support this reading from the supply side: multinationals whose production depends heavily on local intermediates are more likely to remain in a disaster-struck host country. The parameter governing that result is the cost share of local inputs in multinational production, which is not the same object as chain position measured by distance from final demand; the two are plausibly correlated but not identical, and the argument is carried over here by analogy. Firm-level evidence for Belgium following the 2021 floods (Bijnens et al., 2024) points in a compatible direction, with persistent losses concentrated among firms whose suppliers were affected.

2.5.4 From the literature to RQ3

Upstream position determines asset specificity and the systemic risk carried by the node, and therefore the multinational's incentive to retain rather than close it after a shock. On this reasoning, historical exposure would amplify the retaining effect: where events recur, upstream affiliates are more likely to have been built with operational redundancy, lowering the cost of continuing relative to exit. For downstream affiliates the same reasoning implies limited moderating power, since relocation remains low-cost regardless of preparedness.

RQ3: Does a host country-sector's position in the global value chain moderate the relationship between natural disasters and subsequent affiliate presence, and is any such moderation itself conditioned by the host country's historical exposure?

2.6 Synthesis

Two gaps follow from this review. First, none of these literatures varies hazard type, host-country institutional environment and value-chain position within a single empirical framework; each holds two of the three fixed while examining the remaining one. Second, the null result of Oh & Oetzel (2011) at the affiliate level has not been re-examined across a substantially broader set of host countries and industries, despite being the finding most directly comparable to the object of this thesis.

Table 2-1: Research questions, theoretical anchors and principal references

Research Question	Theoretical anchor	Key references
RQ1	OLI: entry margin	Kato & Okubo (2022); Oh & Oetzel (2011)
RQ2	Post-entry adjustment	Neise et al. (2022); Kato & Okubo (2022, §5.2); Hsiang & Jina (2014)
RQ3	GVC: propagation and retention	Altay & Ramirez (2010); Barrot & Sauvagnat (2016); Antràs & Chor (2022); Kato & Okubo (2022)

3 METHODOLOGY

3.1 Data Sources and Sample Construction

This section describes the five data sources underlying the empirical analysis, the cleaning and harmonisation decisions applied to each, and the construction of the final estimation dataset. All sources are merged at the level of the (year, home_country, host_country, industry) dyad-year cell, yielding a balanced panel spanning 2008 to 2019.

3.1.1 Emergency Events Database (EM-DAT)

EM-DAT is a global database of natural and technological disasters maintained since 1988 by the Centre for Research on the Epidemiology of Disasters (CRED) at UCLouvain, and records over 26,000 events since 1900. Its main purpose is to aid humanitarian efforts by offering standardized, internationally comparable data on disaster events and their human impacts. For an event to enter the EM-DAT dataset, at least one of the following threshold criteria must be met:

- At least 10 people dead or missing
- At least 100 people injured, affected, or left homeless
- Call for international assistance
- Declaration of state of emergency

The EM-DAT database compiles disaster reports from diverse sources such as UN agencies, national emergency authorities, the International Federation of Red Cross and Red Crescent Societies, academic institutions, insurance reports, and press wire services. The key variable analysed here is Total Affected, which sums injured, homeless, and directly affected individuals.
Research by (Delforge et al., 2025) identifies four systematic biases in EM-DAT data:

- **Time bias**: Reporting has increased over time, making recent disaster frequency appear higher.
- **Geography bias**: Reporting quality and completeness vary widely between countries.
- **Hazard-related bias**: Slow-onset disasters like droughts and sea-level rise are under-represented compared to sudden events.
- **Threshold bias**: A minimum severity threshold causes inherent censoring.

One of the objectives of this research is to document the geography bias more rigorously, in order to characterise and, where possible, correct for its influence on the future analysis. We link the number of EM-DAT events reported per country over the 1988–2019 period to each country's governance effectiveness (see Section 3.1.3), The assumption is that countries with stronger institutions have better disaster-reporting agencies. Figure 3-1 shows EM-DAT event frequency per million inhabitants plotted against the 2008 Government Effectiveness estimate, with Tuvalu excluded as a micro-state outlier.

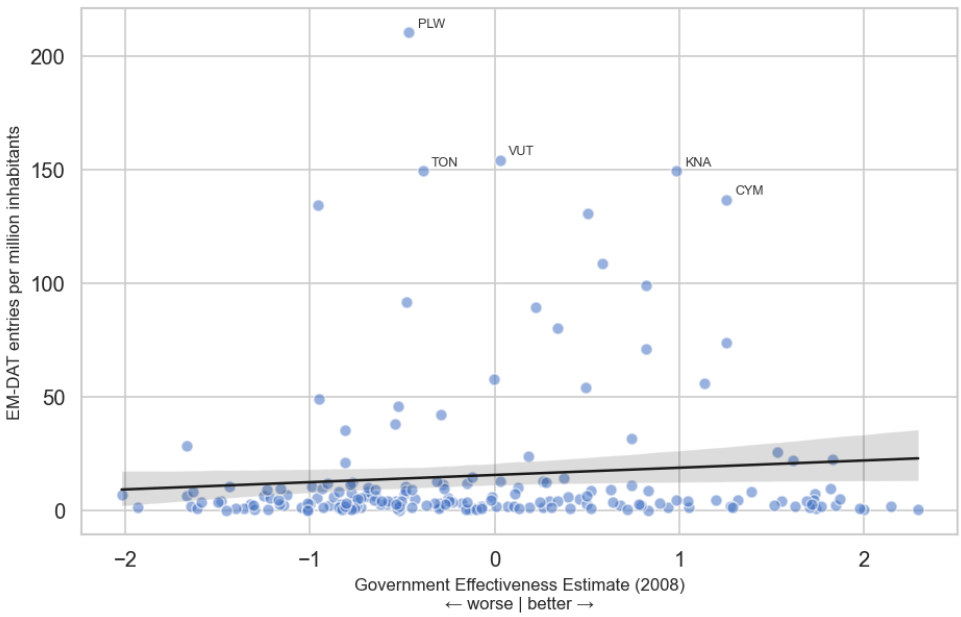


Figure 3-1: Per-capita EM-DAT Reporting Frequency (1988-2019) vs Government Effectiveness (2008)

The positive association confirms the expected pattern: better-governed countries tend to report more events. This raises a further concern: that worse-governed countries may in fact record only the most extreme events, while better-governed countries record the full distribution including moderate ones. To test this, we define disaster severity at the event level as the log of per-capita affected:

$$Disaster\ severity\ = ln(1\ +\ sum(total_affected)\ /\ pop_panel_year) \quad (1)$$

This log-hyperbolic transformation compresses in (1) the skewed distribution of affected counts while preserving the full ordering of intensity and retaining observations with zero affected. Figure 3-2 plots the median per-event severity against governance effectiveness.

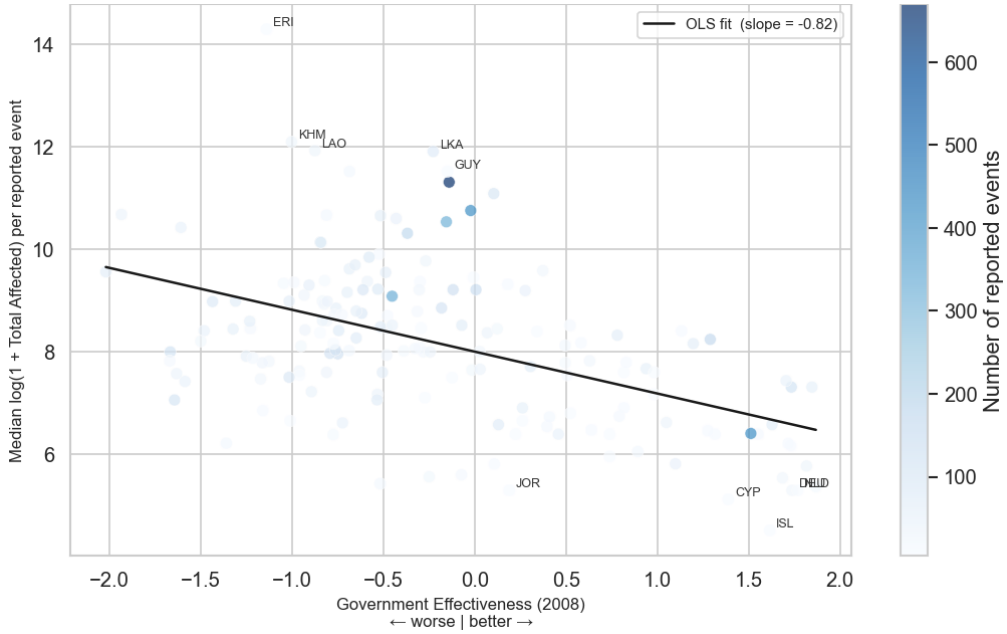


Figure 3-2: Per-event median severity vs Governance Effectiveness

The negative slope confirms the severity-bias conjecture: events recorded in worse-governed countries tend to be substantially more severe on average, consistent with a reporting system that captures only catastrophic events and omits moderate ones.

For the purposes of the subsequent empirical analysis, the 13 natural disaster types present in EM-DAT are consolidated into three groups. The consolidation follows the hazard families of the CRED–Munich Re classification that EM-DAT itself applies (Below et al., 2009), and the decision to treat those families separately rather than jointly follows Altay & Ramirez (2010), who find that damage from windstorms and floods behaves differently enough from damage from earthquakes to constitute evidence against an all-hazards approach:

- ***Sudden physical shocks*** (earthquakes, volcanic activity, dry and wet mass movements, meteorite impacts): geophysical events with no seasonal predictability.
- ***Seasonal physical shocks*** (floods, storms, wildfires): hydro-meteorological events that recur with broadly predictable seasonal timing.
- ***Slow-onset climatic shocks*** (droughts, extreme temperatures): gradual-onset phenomena that build over weeks or months.

The distinction that matters for the empirical design is anticipation. An earthquake can occur in any month of any year, whereas floods and storms concentrate in seasons whose timing is known in advance, monsoon flooding being the clearest case; the first group is accordingly treated as approximately unexpected and the second as approximately expected. The approximation fails in both directions, and the failures are not symmetric. The seasonal group contains events no one expected — the floods of July 2021 in Belgium and western Germany had no recent precedent in the region — while the sudden group contains events that were partly foreseen, since volcanic activity is often forecast by geological monitoring in the weeks preceding an eruption. The second error is judged the smaller: geophysical hazards lack the seasonal regularity that makes anticipation routine, whereas the seasonal group mixes recurrent monsoon flooding with events unprecedented for the regions they struck. A warming climate compounds the asymmetry, since hazards that would have been unexpected two decades ago are entering the expected distribution in regions that had not previously faced them. Resolving this would require classifying events rather than hazard families, which is not feasible on the present data and is taken up in Chapter 5.
For each group, a country-year severity variable is computed as in (2):

$$sev_{group,year} = \ln\left(1 + \frac{\sum\left(total\ affected\ per\ event_{group,year}\right)}{population_{year}}\right) \tag{2}$$

Long-run disaster proneness is measured by event frequency rather than by severity. For each country, the number of EM-DAT events recorded over a fixed window is divided by host population expressed in millions, giving a count of reported events per million inhabitants:

$$hist_cnt_static = events[1987,2007] / \left(\frac{pop_{1987}}{1{,}000{,}000}\right) \tag{3}$$

$$hist_cnt_dynamic_{year} = events[year-21, year-1] / \left(\frac{pop_{year-21}}{1{,}000{,}000}\right) \tag{4}$$

Both measures are drawn from EM-DAT events recorded from 1987 onward, using World Bank population data as the denominator. A country with no EM-DAT entry in the relevant window receives a value of zero after imputation in the estimation sample; a country absent from the World Bank population data receives a missing value that triggers listwise deletion.

Frequency is preferred to severity for this purpose because severity aggregated over a two-decade window is dominated by whichever single event in it was largest, which is precisely the quantity that the inclusion thresholds and reporting differences documented above distort most, whereas a count uses only what the database is designed to establish: that an event crossed the threshold.

3.1.2 Foreign Affiliates Statistics (FATS) – Eurostat

Eurostat compiles and publishes the Foreign Affiliates Statistics (FATS) Outward database, which annually reports on foreign affiliates controlled by enterprises headquartered in EU Member States and selected associated countries (Eurostat, 2025). A foreign affiliate is defined as a company where the parent holds over 50% of voting power, either directly or indirectly, granting operational control. This includes wholly-owned subsidiaries and majority joint ventures but excludes minority and non-controlling stakes.

For each observation identified by (reporting country × partner entity × industry × year), three key variables are extracted: number of affiliates (ENT), employees (EMP), and turnover in thousands of euros (TUR). Other indicators are excluded. The dataset covers 29 home countries (EU-27, UK, and Norway) and 236 host entities, including sovereign states and some non-sovereign territories. Pre-aggregated geographic groups like WORLD or EU28 are excluded as they lack specific disaster observations and cannot be matched with control variables.

The panel spans 2008 to 2019. Eurostat applies confidentiality rules: if a cell contains only 1 or 2 affiliates, values are suppressed (marked missing) rather than shown as zero, preserving the distinction between no activity and censored data. This affects EMP and TUR variables more than ENT, with missing data rates of 12.9% for ENT, 20.5% for EMP, and 21.8% for TUR.

Affiliate activity is classified by industry using the NACE Rev.2 system — the European standard for statistical classification of economic activities —, which segments economic activities into 21 broad sections (A–U). To simplify analysis, only section-level data are used, dropping finer division codes (such as C10, C13, or C10-C12), resulting in 16 sections (A–S) in the estimation sample after excluding sections lacking meaningful control or not covered by FATS Outward's business-economy scope. Details on section aggregation and harmonization with OECD TiVA classification are in Appendix A.

FATS uses a dual-structure due to host size differences: for 60 major partner economies, section-level data are published, allowing industry-specific and global value chain (GVC) analysis. For smaller or less active hosts, only cross-sector aggregates (code B-S_X_O) are available, capturing activity without industry breakdown.

Two empirical strategies arise:

- Strategy A uses the aggregate B-S_X_O rows, maximizing geographic coverage (up to 185 host entities) but ignoring industry differences (used for hypotheses RQ1 and RQ2).
- Strategy B uses section-level rows (A–S) for about 60 host countries, enabling merge with TiVA GVC data (used for RQ3).

These strategies are mutually exclusive in regressions to avoid duplicate counting. Section K (financial and insurance) is excluded due to distortions from shell companies without operational staff.

A notable feature of the underlying affiliate activity is the high proportion of inactive home-host pairs. As detailed in Section 3.1.6, the constructed extensive-margin sample — the full cross-product of the 5,481 home-host pairs observed over the twelve panel years — shows that 54.8% of pairs never record a single affiliate over 2008–2019, and 66.1% of all pair-year cells record zero affiliates.

3.1.3 World Bank Macro-Economic and Governance Indicators

The analysis uses macro-economic and institutional control data from two World Bank datasets. GDP per capita is calculated from the World Development Indicators (WDI) indicator NY.GDP.MKTP.KD (GDP at constant 2015 USD) divided by population data to obtain per-capita values (World Bank, 2024). The constant 2015 USD series is preferred over current USD to ensure comparability across countries and years by removing inflation and exchange rate effects. This variable is included in all panel models in logarithmic form and lagged by one period (t−1) to avoid simultaneity bias, as disasters in year t may reduce GDP in the same year, which would obscure causal effects if current-year GDP were used. Governance is measured by the Government Effectiveness (GE) estimate from the Worldwide Governance Indicators (Kaufmann et al., 2010), which captures perceptions of public service quality, civil service, and policy implementation on a standardized scale ranging roughly from −2.5 to +2.5. GE is chosen because it best reflects the state's capacity to respond to and manage physical shocks. Like GDP per capita, the GE estimate is lagged by one period in regressions to mitigate potential endogeneity with contemporaneous disasters.
Population is also drawn from the same WDI population series and enters the cross-sectional specification in log form as a standard gravity control for market size.

3.1.4 Trade in Value Added (TiVA) – OECD

The Trade in Value Added (TiVA) database, published by the OECD, provides input-output-based decompositions of bilateral trade flows that allow the domestic and foreign content of exports to be separated.(OECD, 2023) In the context of this thesis, TiVA serves as the source for a measure of each country-sector pair's position in the global value chain (GVC), which is used as an interaction variable in RQ3.
The GVC position index follows Koopman et al. (2014) and is computed as:

$$GVC\ Position = ln(1 + EXGR_{INTDVASH}) - ln(1 + EXGR_{FVASH}) \quad (5)$$

Where $EXGR_{INTDVASH}$ measures the share of domestic value added embodied in exports of intermediate products, reflecting the industry×country's upstream role (*forward linkage*) in global value chains: it represents the locally created value incorporated into the production or exports of direct commercial partners. $EXGR_{FVASH}$ is the share of foreign value added in gross exports (a measure of *backward linkage* or downstreamness). A positive GVC index value indicates a net upstream position: the country-sector primarily sells intermediates to others. A negative value indicates a net downstream position, where the country-sector absorbs foreign-produced inputs.
Coverage is approximately 81 countries, 19 NACE sections (A–S), and 12 years (2008–2019), all benchmarked against a world-aggregate counterpart. The alignment of TiVA codes with the FATS classification used in this study is detailed in Appendix A.

Figure 3-3 shows substantial within-sector dispersion in the GVC position index, alongside the expected between-sector ordering: extractive and heavy-manufacturing sections such as B, N and M skew upstream, while distributive and service sections such as I, Q and R skew downstream. The dispersion is itself informative. NACE sections are broad aggregates, and countries specialising in different sub-activities within the same section can occupy very different points on the chain. Appendix A illustrates this for section B. Because the index is observed only at section level, that composition effect cannot be separated from genuine within-country-sector heterogeneity in the

current data, so the wide boxes for sections such as C, H and L reflect both cross-country differences in specialisation and the coarseness of the aggregation itself.

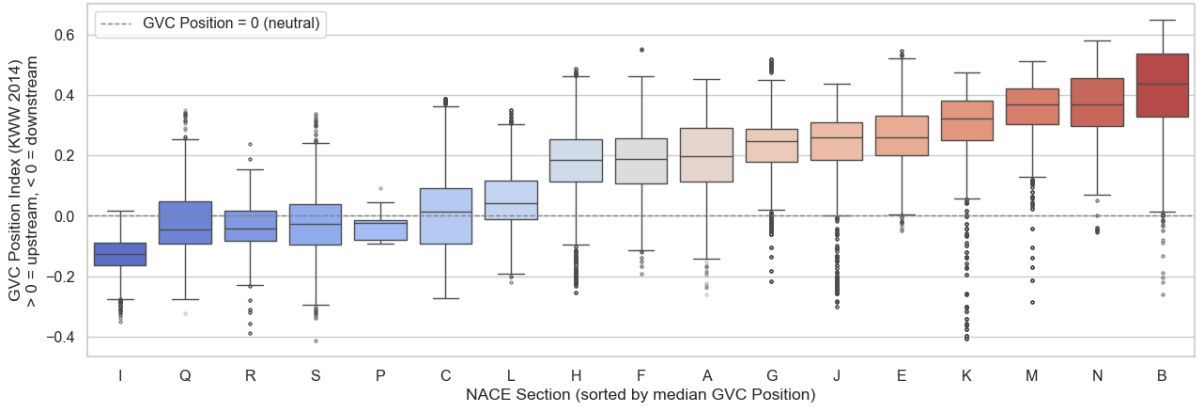


Figure 3-3: GVC Position by NACE section

Note that the description of each industry's section can be found in Appendix A.

3.1.5 CEPII Gravity Dataset

Bilateral gravity controls come from the CEPII dist_cepii dataset (Mayer & Zignago, 2011), which provides time-invariant pair-level data for all country pairs. The retained variables include the population-weighted geodesic distance between the two largest cities (distw, in kilometers), a contiguity indicator (1 if countries share a land border), a common colonial history indicator (1 if a colonial relationship has ever existed), and a common official language indicator. These variables align with the standard structural gravity model and control for natural barriers and channels facilitating foreign direct investment.
Additionally, two time-invariant geographic host-country variables complement the time-varying controls: land surface area (in square kilometers) and a binary landlocked indicator (1 if the country has no coastline).

3.1.6 Panel Construction

3.1.6.1 Final Panel Construction

Building the final dataset proceeds in two stages: a merge stage that combines all five sources into a single dyadic panel, and a sample-construction stage that applies the cleaning rules needed for estimation. Keeping the two conceptually separate ensures that every deletion or imputation applied before estimation reflects an explicit, motivated choice, rather than being an implicit by-product of how the sources happened to be combined.

3.1.6.2 Merge structure

The merge uses Eurostat FATS Outward as its structural backbone: every other source is joined onto FATS observations, so a country-pair-industry-year cell that never appears in FATS never appears in the panel. The resulting unit of observation is the cell (year, home country, host country, industry), where industry takes one of three forms: a NACE Rev.2 section letter, or one of the two cross-sector aggregate codes Eurostat reports for host countries where detailed sectoral activity is confidential or simply not published: one covering the whole business economy, the other covering services only. These aggregate rows are the merge-time counterpart of the dual reporting structure described earlier: they extend geographic coverage at the cost of industry detail.

Two additional cleaning decisions are made at this stage. First, wherever every affiliate-count observation feeding a given cell is confidentiality-suppressed, the cell is coded as missing rather than zero, preserving the distinction between "no affiliates" and "value withheld." Second, 9.7% of the merged panel report a positive affiliate count together with exactly zero employees. This pattern is concentrated among small-affiliate-count cells (ENT ≤ 3 for 61% of these cells, against 7% in cells where employment is genuinely nonzero), consistent with a secondary confidentiality threshold on employment that is not always reflected as missing in the same way as the affiliate count; a smaller tail with large affiliate counts and zero employment, concentrated in known holding-company jurisdictions, more plausibly reflects genuine shell structures. In both cases the reported zero is not economically informative, so employment is set to missing rather than retained as a true zero.
This correction, combined with ordinary confidentiality suppression, is the main reason employment and turnover carry substantially higher missing rates than the affiliate count itself.
Onto this backbone, the disaster, historical-exposure, gravity, macroeconomic, governance, global-value-chain, and geographic control variables described in the preceding sections are each joined on their natural key (country-year, country pair, or country-sector-year). The resulting merged panel covers 2008–2019, comprises 29 home countries and up to 185 host countries depending on the control considered, and totals 341,576 dyad-year-industry cells.

3.1.6.3 Constructing the estimation sample

From this merged panel, the sample used in the regression analysis is obtained through a further sequence of cleaning steps, applied in order:

1. Missing disaster and historical-exposure values are set to zero. A missing value in these columns means the country-year has no EM-DAT record at all — economically equivalent to no disaster having occurred — rather than an unobserved quantity.
2. Government effectiveness and GDP per capita are replaced by their one-year-lagged values, for the simultaneity reasons already discussed above, and the raw contemporaneous GDP variable is dropped so that it cannot be reintroduced unintentionally in a later specification.
3. Domestic "investment" rows, where the home and host country coincide, are removed, as they are not meaningful in a cross-border framework.
4. NACE Section K (Financial and Insurance Activities) is excluded: its affiliate counts are dominated by holding structures, and its global-value-chain position is mechanically upstream for reasons unrelated to physical trade exposure.
5. Rows missing the host-country macroeconomic controls or the bilateral distance variable are dropped. These are almost always small territories or statistical residuals with no coverage in the underlying sources, rather than a random subset of the panel.
6. Any (home, host, industry) triplet that never records a positive affiliate count over the full 2008–2019 window is removed, since it contributes no meaningful variation to a model of affiliate activity levels.

After these steps, the estimation sample contains 104,453 observations: 66,949 at the section level (16 NACE sections), 26,000 at the whole-business-economy aggregate level, and 11,504 at the services-only aggregate level. It spans 29 home and 185 host countries across 10,754 distinct home-host-industry triplets. Remaining missingness reflects the underlying sources rather than the construction process itself: employment and turnover are still missing for 28.9% and 31.6% of rows respectively (confidentiality suppression plus the shell-entity correction described above), and the global-value-chain position index is missing for 36.8% of rows, since it is only ever observed for section-level rows in host countries covered by the OECD value-added trade database, and is absent by construction for both aggregate industry codes. Because the pruning step above conditions on a triplet ever having recorded activity, the affiliate count is heavily concentrated on positive values: 79.5% of rows record a positive count, 18.8% record a genuine within-triplet zero, and 1.7% remain missing under confidentiality suppression.

3.1.6.4 An extensive-margin sample

The sample above is well suited to studying how already-established affiliates respond to shocks, but by construction it is silent on market entry: any home-host-industry triplet that never appeared in the source data with a positive affiliate count has already been removed. A second, complementary sample is built specifically to study this margin. Rather than pruning inactive triplets, it reinstates them deliberately, as the full cross-product of every home-host country pair with usable macroeconomic, governance, and gravity data (5,481 pairs) against the twelve panel years, restricted to the whole-business-economy aggregate industry level. The aggregate level is used deliberately rather than section-level detail, because market entry is naturally a country-pair decision: a firm can be present in one sector and absent from another within the same host country, which would make a section-level "entry" indicator ambiguous.

For each pair-year, affiliate counts are filled in from the source data where available. Pairs with no matching record are coded as true structural zeros (no affiliates, no employees, no turnover), while pairs that do appear in the source data but under confidentiality suppression retain a missing value rather than being coded as zero, preserving the same distinction used in the main sample. A companion indicator flags whether a given home-host pair ever recorded a positive affiliate count in any year of the panel, separating pairs with observed activity at some point (45.2% of rows) from pairs that are inactive throughout (54.8%). All other controls and lags are constructed exactly as in the main estimation sample, but no activity-based pruning is applied, since retaining the zeros is the entire purpose of this sample.

The result is a balanced panel of 65,772 observations (5,481 pairs × 12 years), of which 66.1% record zero affiliates, 33.1% record a positive count, and 0.8% remain missing.

3.2 Common Econometric Identification Choices

Across all three empirical strategies, four identification choices recur — though, as detailed below, their concrete implementation differs between the cross-sectional design (RQ1) and the panel designs (RQ2, RQ3). Table 3-1 summarises how each choice maps onto each specification.

3.2.1 Time-invariant controls: direct inclusion versus fixed-effects absorption

The bilateral gravity and geography variables — distw, contig, colony, comlang_off, landlocked_host, surface_area_host — are constant within a given home–host pair across the panel window. In the cross-sectional specification (RQ1), where each dyad contributes a single observation, these variables carry the identifying variation and are included directly, following standard structural gravity practice (Head & Mayer, 2014). In the panel specifications (RQ2, RQ3), a dyad fixed effect α_{ij} is included to absorb all time-invariant dyad characteristics. Because a variable constant within ij lies exactly in the column space spanned by the dyad-fixed-effect dummies, it is perfectly collinear with α_{ij} and cannot be separately identified within the panel estimator; it is therefore mechanically dropped rather than omitted by choice. Host- and home-country population (pop_host, pop_home) follow a related but distinct logic: they vary from year to year, so they are not exactly collinear with α_{ij}, but their within-dyad annual variation is small relative to their between-dyad variation, and — once a year fixed effect γ_t is also included — this residual variation becomes quasi-collinear with γ_t, producing unstable coefficient estimates. Population is therefore excluded from the panel specifications by deliberate choice rather than mechanical necessity, and instead enters only indirectly, as the denominator used to construct the per-capita historical exposure measures (Section 3.1).

3.2.2 One-period lag on macroeconomic and governance controls

Host- and home-country GDP per capita (ln_gdp_pc_host, ln_gdp_pc_home) and government effectiveness (ge_host, ge_home) enter every specification at $t-1$ rather than contemporaneously. A disaster occurring in year t can simultaneously depress measured GDP and perceived government effectiveness in that same year; including their contemporaneous values would therefore constitute a bad control (Angrist & Pischke, 2009) — a variable itself affected by the treatment — and would partially absorb the very shock effect the model is designed to estimate. Lagging both controls by one period breaks this channel of simultaneity while preserving their role as predetermined confounders of MNE location and activity decisions.

3.2.3 Mean-centring of interaction terms

All continuous variables entering an interaction term (ge_host, gvc_position, ln_hist_cnt_pc) are centred on their estimation-sample mean before the interaction is formed. This serves two purposes. First, it removes the structural, non-substantive correlation between an interaction term and its constituent main effects that arises purely from scale, which otherwise inflates variance inflation factors without reflecting genuine multicollinearity in the underlying relationship (Iacobucci et al., 2016). Second, it renders the main-effect coefficients directly interpretable as the effect of one variable evaluated at the sample-average level of its interaction partner, rather than at the (frequently meaningless) value of zero.

3.2.4 Strictly pre-period historical exposure windows

The historical disaster-exposure measures (hist_cnt_static, hist_cnt_total_dynamic) are constructed so that no window ever extends into the contemporaneous or future panel year, guaranteeing that they are predetermined with respect to the shock being studied. Two variants are used, and the choice between them follows directly from the fixed-effects logic above rather than being interchangeable. The static measure, covering the fixed window 1987–2007, is time-invariant per country and is used in the cross-sectional specification (RQ1), where no dyad fixed effect exists to force it out. The dynamic measure, covering the rolling window $[t-21\,,t-1]$, is used exclusively in the panel specifications (RQ2, RQ3): unlike the static measure, it varies year to year, so it survives the double-demeaning within-transformation described in Section 3.3.2 and retains identifying variation net of the dyad fixed effect.

3.3 Estimation Methods

Table 3-1 maps each research question onto its sample, temporal design, and distributional choice; the subsections that follow present the mechanics of each building block in turn.

Table 3-1: Empirical specifications by research question

	RQ1	RQ2	RQ3
Sample / strategy	2008–2019 cross-section (pair average), Strategy A	Panel, Strategy A	Panel, Strategy B (section-level, merged with TiVA)
Temporal design	Static (one observation per pair)	Local Projections, $T = 0 \rightarrow T = 5$	Local Projections, $T = 0 \rightarrow T = 3$
Distributional choice	OLS only	Log-OLS[1]and PPML, estimated at every horizon	Log-OLS[1] and PPML, estimated at every horizon
Gravity / geography controls[2]	Included directly	Absorbed by dyad fixed effect	Absorbed by dyad fixed effect
Population	Included directly	Excluded (see Section 3.2)	Excluded (see Section 3.2)
Historical exposure window	Static, 1987–2007	Dynamic, $[t-21, t-1]$	Dynamic, $[t-21, t-1]$
Interaction terms	Mean-centred	Mean-centred	Mean-centred (triple interaction)
Standard errors	Heteroskedasticity-robust (HC3)	Clustered by dyad	Clustered by dyad

3.3.1 Cross-Sectional OLS Baseline (RQ1)

RQ1 is estimated by ordinary least squares on the 2008–2019 pair-level cross-section, with heteroskedasticity-robust (HC3) standard errors (MacKinnon & White, 1985). HC3 is preferred over the classical or HC1 estimator because the cross-section is unbalanced across host countries of very different sizes, and HC3's leverage-based weighting is more conservative in finite samples with high-leverage observations, a relevant concern here given the presence of small, low-observation host economies alongside major trading partners. No fixed effects are applicable at this stage, since the collapse to one observation per dyad leaves no within-dyad variation to exploit; this is precisely the estimator that Section 3.2 contrasts with the panel specifications below.

3.3.2 High-Dimensional Two-Way Fixed Effects (TWFE)

The panel specifications for RQ2 and RQ3 include a dyad fixed effect α_{ij} and a year fixed effect γ_t:

$$Y_{ij,t} = \beta \times Shock_{ij,t} + \theta \times Controls_{ij,t-1} + \alpha_{ij} + \gamma_t + \varepsilon_{ij,t} \quad (6)$$

With thousands of dyads and twelve panel years, explicitly constructing dummy variables for every fixed effect is computationally impractical. Following Gaure (2010), the linear model is instead estimated via the within transformation (double demeaning): each variable is swept of its dyad mean and its year mean in alternation,

$$\tilde{x}_{ij,t} = x_{ij,t} - \bar{x}_{ij\cdot} - \bar{x}_{\cdot\cdot t} + \bar{x}_{\cdots} \quad (7)$$

[1] Log-OLS is used as a robustness check in this case, PPML remains the main model.

[2] (distw, contig, colony, comlang_off, landlocked_host, surface_area_host)

and the model is estimated by OLS on the demeaned variables, which is numerically equivalent to including the full set of dummies but at a fraction of the computational cost.

3.3.3 Poisson Pseudo-Maximum Likelihood (PPML)

The log-OLS estimator has two well-known limitations in this setting. First, it mechanically discards every dyad-year with zero affiliates, employees, or turnover — precisely the observations most informative about complete exit — introducing a selection bias against the cells that record complete exit. Second, under multiplicative heteroskedasticity, common in trade and FDI flows where the variance of the error term scales with the size of the flow, OLS estimated on a log-transformed dependent variable is inconsistent, because $E[\ln y] \neq \ln E[y]$ under heteroskedastic errors. Santos Silva & Tenreyro (2006) show that the Poisson pseudo-maximum-likelihood estimator, which models the conditional mean directly rather than its log, remains consistent under any form of heteroskedasticity and naturally accommodates zero-valued outcomes.
PPML assumes an exponential conditional mean and is estimated by iteratively reweighted least squares. Within each iteration the dyad and year fixed effects are absorbed by the alternating-projections algorithm of Gaure (2010), as implemented in pyfixest's fepois. Appendix B reports the estimating equations and the convergence criterion.

3.3.4 Dynamic Identification via Local Projections

To trace the effect of a disaster shock beyond its year of occurrence, RQ2 and RQ3 use the Local Projections method of Jordà (2005) rather than a global dynamic model such as a Vector autoregression (VAR). For each horizon $h \in \{0, \dots, H\}$, a separate regression is estimated:

$$Y_{ij,t+h} = \beta_h \times Shock_{ij,t} + \theta_h \times Controls_{ij,t-1} + \alpha_{ij} + \gamma_t + \varepsilon_{ij,t+h} \quad (8)$$

using the TWFE (Section 3.3.2) or PPML (Section 3.3.3) machinery described above, applied independently at each h. No lag is imposed on the shock variable itself; instead, the dependent variable is shifted forward within each dyad (df.groupby('dyad_id')[y].shift(-h)), so that β_h traces the impulse response of Y to a shock at t, h periods later. Relative to a VAR, Local Projections impose no restriction on the shape of the dynamic response, accommodate the non-linear PPML link and the higher-order interaction terms required by RQ3 without additional derivation, and are robust to misspecification of the underlying dynamic process (Jordà, 2005).
Two limitations follow directly from this design and are addressed explicitly. First, projecting the dependent variable forward mechanically induces serial correlation in the error term across overlapping horizons for a given dyad; standard errors are therefore clustered at the dyad level (cluster = dyad_id) throughout, following standard practice for panel data with a moderate number of clusters (Cameron, C., & Miller, D., 2015). Second, the estimation sample shrinks as h increases, since dyad-years near the end of the panel (e.g., a 2018 shock) have no realised outcome at $T + 5$; RQ2's $T + 5$ horizon is therefore identified from a strictly earlier cohort of shocks (2008–2014) than $T = 0$, a composition change that is checked explicitly via a cohort-restricted robustness specification (Section 4.3).

4 RESULTS AND CRITICAL DISCUSSION OF RESULTS

4.1 Reading the results

This chapter reports the estimates obtained for the three research questions formulated in Chapter 2 and discusses them critically in light of what the data are able to identify. Each question is treated in a section of its own — RQ1 in Section 4.2, RQ2 in Section 4.3, RQ3 in Section 4.4 — and each follows the same order: the estimates of the preferred specification, the robustness checks, and the substantive discussion. Section 4.5 then turns to three features of the data that bear on all three sets of estimates and are therefore treated jointly rather than restated in each section, and Section 4.6 summarises the answer reached on each.

Each question is answered by a specific coefficient, and identifying which one in advance is what keeps the discussion that follows from ranging over whichever estimate happens to be significant. RQ1 turns on the interaction between government effectiveness and historical disaster frequency, read together with the frequency term it conditions: the pair describes whether the association between exposure and affiliate presence varies with institutional quality, and in which direction. RQ2 turns not on a coefficient but on a sequence: the six horizon-specific estimates of the sudden-shock term, and their counterparts on the interaction with historical exposure, which together trace how presence evolves after a shock and whether that path differs by exposure history. RQ3 turns on the triple interaction between disaster severity, value-chain position and historical exposure; the lower-order terms are included so that the triple interaction is defined, and are not themselves the object of the question.

Because the three questions ask whether an association exists and in which direction, rather than whether a stated sign is confirmed, the criterion applied throughout is symmetric with respect to sign. An affirmative answer requires an estimate distinguishable from zero whose sign is stable across horizons, estimators and dependent variables. A negative answer requires estimates close to zero and precise enough to exclude the magnitudes at issue. Estimates that satisfy neither condition — imprecise enough to remain compatible with effects of opposite sign — answer the question in neither direction, and are reported as such rather than assimilated to either.
For RQ2 the literature supplies a reference trajectory rather than an expectation. The creative-destruction account reviewed in Section 2.4, were it to operate on this margin, would produce a response that is negative on impact and positive three to five years later; that pattern is used here as a benchmark against which the estimated sequence is compared, not as a prediction this thesis adopts. No horizon is privileged within the window, and where the estimated sequence does not trace the benchmark, the comparison is reported on the sign pattern alone and the significance of any individual horizon does not arise.

Four reporting conventions apply throughout.
First, the reported specification for each question is presented in full, so that its focal coefficients can be read against the controls that accompany them. The robustness specifications are presented through their focal coefficients alone; the complete models are in Appendix D to Appendix F.
Second, for RQ2 and RQ3 the reported specification is PPML. The two estimators do not estimate the same quantity: log-OLS conditions on cells recording a strictly positive outcome and describes adjustment among cells that remain active, whereas PPML retains the cells that fall to zero and describes presence inclusive of its disappearance. Both questions are stated over presence, and RQ3 over retention specifically, so the second quantity is the one they ask about. The log-OLS estimates are accordingly reported as a robustness check, establishing whether an answer obtained on the full sample also holds among the cells that remain active. Third, a robustness specification is discussed

in the body only where it diverges from the reported one. Divergence rather than agreement is what carries information here.
The third convention matters for everything that follows. The design produces far more focal estimates than it can support individually. Crossing three focal coefficients with six horizons, two estimators and two dependent variables yields seventy-two estimates for RQ2 alone, and the cohort-restricted specifications double that count. At a five per cent threshold, roughly seven significant coefficients are expected from the number of tests alone, with no effect present anywhere in the design. No correction for multiple testing is applied, and none would restore power at this scale. Instead, no coefficient is interpreted on the strength of its own p-value: interpretive weight is given only to patterns that hold across horizons, estimators and dependent variables, and the seasonal-shock placebo, predicted null by construction, is used throughout as a running check on how often significance arises by chance.
A final convention concerns null results, which account for most of what this chapter reports. A coefficient that fails to reach conventional significance may be precisely estimated and close to zero, or too imprecise to discriminate between an effect and its opposite; these are different findings and are not described in the same terms here. Where an estimate is imprecise, the confidence interval is reported and the range of effects it fails to exclude is stated explicitly, rather than the result being called an absence of effect. The variation available to identify each coefficient is documented in Section 4.5.

4.2 Ex-ante Location Choice (RQ1)

RQ1 asks whether host countries with a longer record of natural disasters host more or fewer European affiliates, and whether the answer depends on how effectively those countries are governed. Equation (9) answers it on a cross-section of home–host pairs, the interaction term carrying the second half of the question. The estimates are reported first, the specifications that test their stability follow, and the section closes with what the two can and cannot be taken to show.

$$ln(\overline{ENT}_{ij}) = \alpha + \beta_1 \widetilde{ge}_{host} + \beta_2 \, ln\left(\frac{\widetilde{HistCnt}}{pop/10^6}\right)_{host} + \beta_3 \left[\widetilde{ge}_{host} \times ln\left(\frac{\widetilde{HistCnt}}{pop/10^6}\right)_{host}\right] + X_{macro}\gamma + Z_{gravity+geo}\delta + \epsilon \quad (9)$$

4.2.1 Estimates

Table 4-1 reports the estimates. The specification is estimated in three steps. Model 1 contains government effectiveness together with the macroeconomic and gravity controls, and establishes the baseline against which the disaster variables are read. Model 2 adds historical disaster frequency. Model 3 adds its interaction with government effectiveness and is the specification on which the answer to RQ1 rests. The sequence serves a diagnostic purpose: because the gravity coefficients are almost unchanged across the three columns, the exposure variables are not absorbing variation that a mis-specified gravity block would otherwise have left unexplained.
The non-focal part of Model 3 behaves as the gravity literature implies. Distance enters at −0.558, contiguity at +1.119, colonial history at +0.729 and common language at +0.256; host and home GDP per capita and population all enter positively, and landlocked hosts record fewer affiliates. Host surface area is indistinguishable from zero, which is what one expects once population and income are controlled. These magnitudes lie in the range that structural gravity estimation routinely produces (Head & Mayer, 2014), and they matter here in one respect only: the focal coefficients are read off a specification whose remaining parameters are not anomalous. Turning to those coefficients, historical disaster frequency enters positively and is significant at the one per cent level in both Model 2 (0.242) and Model 3 (0.278). Its interaction with government effectiveness enters negatively at −0.137,

Table 4-1: Cross-Sectional OLS Baseline Estimates. Dependent Variable: Log Average Enterprises (2008–2019)

Variables	Model 1	Model 2	Model 3
Institutional & Disaster Variables			
Gov. effectiveness (host, centered)	0.3498***	0.3129***	0.2615***
	(0.0000)	(0.0000)	(0.0000)
Hist. disaster frequency (centered)	—	0.2423***	0.2784***
		(0.0000)	(0.0000)
Gov. eff. × Hist. disaster freq.	—	—	-0.1369***
			(0.0005)
Macroeconomic & Gravity Controls			
Log GDP per capita (host, t−1)	0.3051***	0.3327***	0.3696***
	(0.0000)	(0.0000)	(0.0000)
Log Population (host)	0.3617***	0.4422***	0.4525***
	(0.0000)	(0.0000)	(0.0000)
Log GDP per capita (home, t−1)	1.0506***	1.0663***	1.0705***
	(0.0000)	(0.0000)	(0.0000)
Log Population (home)	0.5932***	0.5962***	0.6000***
	(0.0000)	(0.0000)	(0.0000)
Log Distance	-0.4993***	-0.5549***	-0.5581***
	(0.0000)	(0.0000)	(0.0000)
Contiguity (dummy)	1.2225***	1.1381***	1.1192***
	(0.0000)	(0.0000)	(0.0000)
Colonial history (dummy)	0.7309***	0.7455***	0.7289***
	(0.0000)	(0.0000)	(0.0000)
Common language (dummy)	0.2769**	0.2438**	0.2557**
	(0.0146)	(0.0314)	(0.0241)
Landlocked host (dummy)	-0.2543***	-0.2516***	-0.2180***
	(0.0000)	(0.0000)	(0.0001)
Log Surface area (host)	-0.0015	-0.00074	-0.0052
	(0.9289)	(0.6527)	(0.7508)
Constant	-22.7654***	-24.0036***	-24.6091***
	(0.0000)	(0.0000)	(0.0000)
Observations	2,472	2,472	2,472
Adjusted R2	0.6098	0.6166	0.6192

Notes: Heteroskedasticity-robust (HC3) p-values in parentheses.
**** p<0.01, ** p<0.05, * p<0.1. All continuous interaction variables are mean-centered.*

also significant at one per cent. Since both the dependent variable and the exposure measure are in logs, the coefficient is an elasticity: at the sample mean of government effectiveness, a ten per cent increase in the number of reported events per million inhabitants is associated with a little under three per cent higher average affiliate count.

The interaction qualifies that elasticity rather than reversing it. The marginal association implied by Model 3 is 0.278 − 0.137 × GE, with government effectiveness entering centred, so that the association declines as governance improves and reaches zero at a centred value of approximately +2.03. Given a sample mean government-effectiveness score of +0.30, that corresponds to a raw index value of about +2.34, which no host country in the sample attains: the best-governed host over

the panel window records +2.20. The marginal association therefore remains positive across the entire observed range of governance, falling from roughly +0.60 among the least well-governed hosts to +0.02 among the best-governed. The interaction attenuates the association to something indistinguishable from zero at the top of the observed range without reversing it anywhere within that range, and the reversal implied by extrapolating a linear interaction term lies outside the support of the data.

Two further features of the table bear on what follows. First, the coefficient on government effectiveness falls from 0.350 in Model 1 to 0.313 in Model 2 and 0.262 in Model 3. Roughly a quarter of what the baseline attributes to institutional quality is therefore shared with reported disaster frequency, which is the in-sample counterpart of the association documented in Figure 3.1-1: better-governed countries report more events. Section 4.2.3 returns to what this implies for the reading of the focal coefficients. Second, the explanatory contribution of the disaster block is modest. Adjusted R^2 rises from 0.610 to 0.617 with the addition of historical frequency and to 0.619 with the interaction — a gain of under one percentage point on a specification that gravity and market size already explain well. The exposure variables are statistically well determined and substantively secondary.

4.2.2 Robustness

Table 4-2: Cross-Sectional Robustness Estimates

	Gov. effectiveness (host, centered)	Hist. disaster frequency (centered)	Gov. eff. × Hist. disaster freq.
average EMP	0.2573**	0.3109***	-0.2310***
	(0.0242)	(0.0003)	(0.0053)
ever_active LPM	0.0493***	0.0334***	-0.0117*
	(0.0000)	(0.0000)	(0.0619)
ever_active logit AME	0.0554***	0.0436***	-0.0098
	(0.0000)	(0.0000)	(0.1383)

Table 4-2 reports three variants testing whether the result depends on how affiliate presence is measured and on the functional form imposed on it. Replacing the affiliate count with average employment changes the outcome from the number of affiliates a host attracts to the scale of activity they represent. Replacing it with the ever-active indicator on the extensive-margin sample of Section 3.1.6.4 changes the question again, from how much presence a host attracts to whether a given home–host pair is ever present at all; that indicator is estimated both as a linear probability model and as a logit, from which average marginal effects are reported.

The signs are identical across all three. Government effectiveness and historical frequency enter positively and at five per cent throughout, and the interaction enters negatively throughout. Magnitudes are not comparable across panels: the employment specification is in logs and its coefficients are elasticities, whereas the two extensive-margin specifications are in probability units, so their smaller values reflect the change of scale and not a weaker association.

The variants do, however, separate the two focal results by strength. The main effect of historical frequency is significant at one per cent in every specification. The interaction is significant at one per cent on employment (−0.231), at ten per cent in the linear probability model (−0.012), and not at conventional levels in the logit average marginal effects (−0.010, $p = 0.138$). Of the two coefficients on which RQ1 turns, the level association is the robust one and the moderation is the weaker, and the discussion that follows weights them accordingly. The complete models are reported in Appendix D.

4.2.3 Discussion

The association runs positive, and it runs positive robustly. Of the two mechanisms set against each other in Section 2.3.5, it is therefore the geographic one that the data favour: hazard-prone sites are disproportionately coastal, deltaic or seismically active, which is to say that they are ports, river mouths and dense agglomerations, and the location advantages that follow are not independent draws alongside the hazard but properties of the same site (Dunning & Lundan, 2008). The forward-looking penalty implied by the risk extension in Kato & Okubo (2022) — a shorter expected duration of presence where the perceived frequency of events is higher — is not what this cross-section shows at this level of observation.

The moderation result is harder to accommodate on the same terms. Section 2.3.5 motivated the interaction on informational grounds: a firm that cannot observe how a location will perform under stress can observe the institutions that will govern warning, response and reconstruction, so strong governance might substitute for the experience the firm lacks and offset an unfavourable exposure history. That reasoning anticipates an interaction that attenuates a penalty. What the estimates report is an interaction that attenuates a premium, and does so most where governance is strongest.

A measurement account fits both results at once, and two features of Table 4-2 are what it predicts. The exposure variable counts events that were reported, and Section 3.1.1 established that reported frequency per capita rises with government effectiveness while median per-event severity falls with it. If part of the historical-frequency coefficient measures reporting capacity rather than hazard, then the governance coefficient should fall once historical frequency is entered, the two variables sharing variance in the sample as Figure 3.1-1 implies they should; it falls by roughly a quarter. And the interaction should take the sign it does, since where governance is weak a reported count is a selective sample of true incidence and informative about a good deal besides, while where governance is strong it is closer to true incidence and carries less of that additional information. The marginal association would then weaken as governance rises, and would do so until it approaches zero among the best-governed hosts, which is what Section 4.2.1 reports. This is a candidate reading rather than an established one; it applies to all three questions and is therefore examined, together with what would be required to separate it from the substantive account, in Section 4.5.1.

Two limits on the scope of the claim close the section. The specification is a cross-section with no exogenous variation in exposure, so what is estimated is a conditional association and not an effect. And, as Section 2.3.5 noted, the location margin is observed here as the distribution of affiliate presence over the panel window, which reflects cumulative entry and survival rather than entry alone. A positive association is therefore equally consistent with entry being unaffected by exposure and exit being lower in exposed hosts, and the design cannot separate the two.

4.3 Post-shock dynamics (RQ2)

RQ2 asks how affiliate presence evolves in the five years following a sudden disaster, and whether that path depends on how often the host country has been struck before. Equation (10) answers it by estimating the same equation separately at each horizon, so that the sequence of coefficients traces the path rather than compressing it into a single number. The estimates are reported first, followed by the specifications that test their stability and by a discussion of what the trajectory does and does not establish.

$$E\left[ENT_{ij,t+k} \middle| \cdot\right] = \exp\big(\alpha_{ij} + \lambda_t + \beta_1 \times Sudden_{host,t} + \beta_2 \times \widetilde{Hist}_{host,t} + \beta_3 \times \left[Sudden_{host,t} \times \widetilde{Hist}_{host,t}\right] + \beta_4 \times Seasonal_{host,t} + X_{macro,t-1}\,\gamma\big) \quad (10)$$

4.3.1 *Estimates*

Seasonal shocks enter the specification as a falsification test. The grouping and its rationale are set out in Section 3.1.1: sudden-onset events are treated as approximately unexpected and seasonal events as approximately expected, with the qualifications recorded there. What the placebo asks is whether coefficients estimated on the first group would also appear on the second. A term returning nothing on events that were broadly anticipated, while the treatment returns something on events that were not, locates the result in the timing of the shock rather than in severity, or in reporting behaviour common to both. One asymmetry should be noted before the estimates: the placebo is estimated on 1,119 shock country-years against 289 for the sudden category, so the falsification test is considerably better powered than the test it is meant to validate.

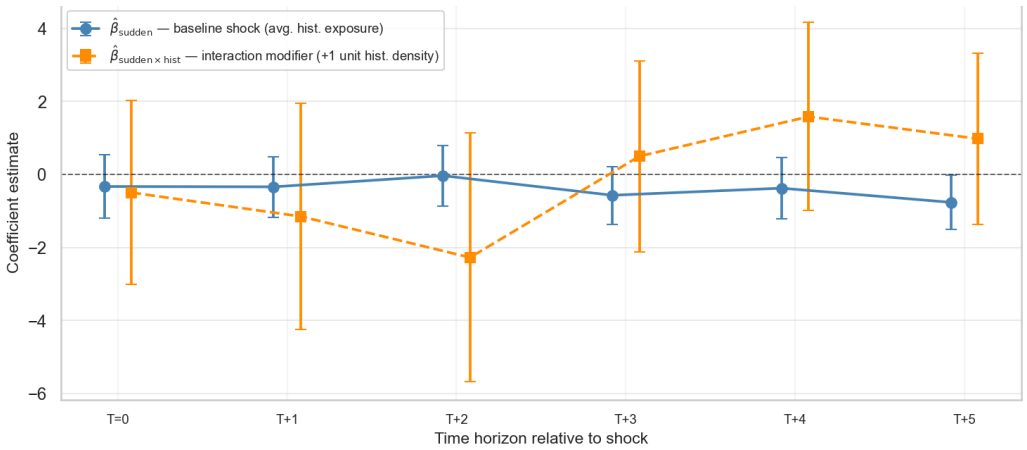


Figure 4-1: Estimated response of affiliate presence to a sudden physical shock, by horizon

Before the coefficients, the variation that identifies them. At h = 0 the estimation sample contains 25,798 dyad-years and 289 distinct country-years in which a sudden physical shock occurred; by h = 5 both have contracted, to 13,770 dyad-years and 168 shock country-years. The contraction is not only a loss of precision. Because a shock requires five subsequent years of observed outcomes to enter the estimate at h = 5, and the panel ends in 2019, the events identifying the longest horizon occurred no later than 2014: they are a subset of those identifying h = 0, and an earlier one. The horizons therefore apply the same design to progressively smaller and progressively older sets of events, which is a reason to read the sequence as a whole rather than to privilege any point within it.
Table 4-3 reports the estimates and Figure 4-1 plots the path. It is negative at every horizon: −0.33 at impact, then −0.34, −0.03, −0.57, −0.38, and −0.77 at five years. Only the last is distinguishable from zero at conventional levels. Read against the benchmark set out in Section 4.1, the comparison fails on the sign pattern before the question of significance arises. The creative-destruction account, were it to operate on this margin, would require the path to turn positive somewhere between three and five years after the event; it does not turn positive anywhere. Whatever else these estimates show, they do not show a rebound.
The interaction with historical exposure is significant at no horizon, and the smallest of its six p-values is 0.190. Nor does it hold a single sign across the window: it runs −0.50, −1.15 and −2.28 over the first three horizons and +0.50, +1.58 and +0.98 over the last three, so the estimated moderation is negative early and positive later, with the crossing falling in the middle of the window. Neither half is distinguishable from zero, and the range the six estimates span is wide relative to the precision with which any of them is measured. The moderation posed in the second half of RQ2 is not identified here in either direction.
The falsification term behaves largely as intended, though not perfectly. Seasonal shocks, which should register nothing if the sudden-shock coefficients capture unanticipated timing rather than

generic severity, return coefficients close to zero at five of the six horizons; at h = 2 the coefficient is +0.21 with a p-value of 0.066. One marginal result in six is close to what the convention of Section 4.1

Table 4-3: Local Projections PPML Event Study Estimates (RQ2) Dependent Variable: Affiliate Count (ENT)

Variables	T=0	T+1	T+2	T+3	T+4	T+5
Disaster Shocks & Interactions						
Sudden physical shock	-0.3315	-0.3404	-0.0349	-0.5711	-0.3773	-0.7654**
	(0.455)	(0.421)	(0.935)	(0.157)	(0.382)	(0.042)
Sudden shock × Hist. disaster freq.	-0.4974	-1.1506	-2.2771	+0.4977	+1.5843	+0.9764
	(0.699)	(0.467)	(0.190)	(0.710)	(0.228)	(0.416)
Seasonal physical shock (placebo)	+0.0548	+0.1490	+0.2117*	+0.1737	+0.1526	-0.0489
	(0.478)	(0.137)	(0.066)	(0.269)	(0.499)	(0.762)
Macroeconomic & governance controls						
Log GDP per capita (host, t-1)	+0.3374***	+0.2667**	+0.1854	-0.0002	-0.1210	-0.2308***
	(0.005)	(0.024)	(0.123)	(0.998)	(0.231)	(0.010)
Log GDP per capita (home, t-1)	+0.9372***	+0.8365***	+0.7213***	+0.3981**	+0.0898	-0.1611
	(0.000)	(0.000)	(0.000)	(0.032)	(0.640)	(0.399)
Gov. effectiveness (host, t-1, centered)	-0.0148	+0.0112	+0.0165	+0.0207	-0.0136	-0.0147
	(0.708)	(0.777)	(0.693)	(0.665)	(0.791)	(0.824)
Hist. disaster frequency (host, dynamic)	-0.0503	-0.0739	-0.0637	-0.0484	-0.0019	-0.0070
	(0.463)	(0.226)	(0.262)	(0.369)	(0.972)	(0.901)
Model Specification & Statistics						
Dyad Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,798	23,343	20,906	18,496	16,110	13,770
Distinct sudden disaster country-years	289	259	237	211	188	168
Distinct seasonal disaster country-years	1,119	1,019	925	830	741	644
Pseudo R^2	0.976	0.977	0.978	0.978	0.979	0.980

*Notes: PPML estimates across Local Projections horizons $T = 0$ to $T + 5$. p-values in parentheses derived from standard errors clustered at the home-host-industry dyad level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Continuous interaction terms are mean-centered.*

leads one to expect from a test designed to find nothing, and Section 4.3.2 shows that it does not survive a change in the estimation sample. It is reported here because a placebo whose single positive is suppressed is not a placebo. That reasoning applies with equal force to the one focal

coefficient in the table that is significant. Across the specifications reported for RQ2 — four estimator–outcome combinations, six horizons, three focal terms, and the cohort restrictions in Appendix E — the design produces well over a hundred focal estimates, of which a handful would be expected to cross the five per cent threshold in the complete absence of any effect. The coefficient at h = 5 is a candidate for being one of them. It is isolated, it appears at the horizon estimated on the smallest and oldest set of events, and it is not accompanied by any pattern in the coefficients surrounding it. Two further observations in Section 4.3.2 support that reading.

Even taken at face value it would not support a quantitative claim. With a coefficient of −0.765 and a p-value of 0.042, the implied ninety-five per cent interval runs from approximately −1.50 to −0.03, which in levels is a decline of anywhere between three and seventy-eight per cent. An estimate that cannot distinguish a negligible change from a collapse is reported as what it is, and nothing in what follows rests on it.

One feature of the control block should be noted so that it is not misread. The macroeconomic and governance controls are dated t−1 relative to the shock while the outcome is dated t+h, so at longer horizons the controls stand increasingly far in time from the quantity being explained. The change in sign of the host GDP per capita coefficient across the horizon range follows from that construction and is not to be read as a structural relationship.

4.3.2 *Robustness*

Table 4-4: Event Study Robustness Summary — Primary Regressors Across Horizons (T=0 to T+5)

Regressors	T=0	T+1	T+2	T+3	T+4	T+5
Panel A: Enterprises (ENT) — OLS						
Sudden physical shock	+0.3528	+0.1702	-0.307	-0.5231	-0.2245	-0.6681*
	(0.301)	(0.650)	(0.390)	(0.404)	(0.598)	(0.095)
Sudden shock × Hist. disaster freq.	-0.1746	-0.2962	-1.131	-0.4686	+0.2986	+0.0492
	(0.890)	(0.776)	(0.180)	(0.713)	(0.823)	(0.957)
Panel B: Employment (EMP) — Log-OLS						
Sudden physical shock	-0.990	-0.673	-0.469	-1.733**	+0.198	+0.703
	(0.352)	(0.547)	(0.476)	(0.036)	(0.771)	(0.287)
Sudden shock × Hist. disaster freq.	+2.376	+1.428	-1.710	+1.910	-1.578	-1.765
	(0.407)	(0.709)	(0.485)	(0.515)	(0.529)	(0.394)
Panel C: Employment (EMP) — PPML						
Sudden physical shock	-0.610	+0.041	-0.368	-0.970*	-0.181	+0.618
	(0.324)	(0.929)	(0.496)	(0.068)	(0.561)	(0.368)
Sudden shock × Hist. disaster freq.	-1.799	-1.185	-1.121	+1.120	+0.336	+3.817***
	(0.325)	(0.548)	(0.556)	(0.485)	(0.830)	(0.002)
Panel D: Entreprises (ENT) – OLS, 2008-2014						
Sudden physical shock	+0.175	-0.150	-0.244	-0.102	+0.102	-0.668*
	(0.662)	(0.664)	(0.594)	(0.858)	(0.836)	(0.095)
Sudden shock × Hist. disaster freq.	+2.531**	+0.797	-0.605	-0.676	-0.143	+0.049
	(0.021)	(0.387)	(0.453)	(0.543)	(0.915)	(0.957)

*Notes: Local Projections estimates across horizons $T = 0$ to $T + 5$. p-values in parentheses derived from standard errors clustered at the dyad level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Macroeconomic controls (ln GDP $pc_{host,\ t-1}$, ln GDP $pc_{home,\ t-1}$, Gov. $Eff._{host,\ t-1}$) are included in all regressions but omitted for brevity.*

Table 4-4 reports three variants and one sample restriction, testing whether the trajectory depends on the estimator, on the outcome measure, or on the composition of the shock cohort. Log-OLS on the affiliate count asks whether the result obtained on the full sample is driven by cells falling to zero rather than by adjustment within the cells that remain active. The two employment specifications ask whether the answer changes when presence is measured by the scale of activity rather than by the number of affiliates. The cohort restriction to shocks occurring between 2008 and 2014, reported in Appendix E, holds the set of identifying events fixed across horizons.

On the first question the answer is that it is not. The log-OLS estimate at h = 5 is −0.67 with a p-value of 0.095: the same sign, a comparable magnitude, and marginal significance on a smaller sample. The negative reading of the total margin at that horizon, such as it is, is not produced by cells leaving the sample.

Beyond that, the variants agree mainly in failing to agree. Across the four estimator–outcome combinations and six horizons, focal significance appears in a handful of cells and never at the same horizon in two of them. Employment estimated by log-OLS returns −1.73 at h = 3 where no other specification returns anything; employment estimated by PPML returns an interaction coefficient of +3.82 at h = 5, a magnitude several times larger than any neighbouring estimate, reproduced nowhere else, at the horizon where the sample is smallest. Under the convention set out in Section 4.1, neither is interpreted on the strength of its own p-value. The dispersion is itself the finding: what varies across specifications here is not the size of an effect but whether one appears at all.

The cohort restriction is instructive in a way that a confirmation would not have been. Holding the shock cohort fixed at events occurring between 2008 and 2014, the impact coefficient moves from −0.33 to −0.06; the placebo, marginally significant at h = 2 in the full sample, returns nothing at any horizon; and the log-OLS interaction at h = 0, which was −0.17 with a p-value of 0.890 in the full sample, becomes +2.53 with a p-value of 0.021. Significance is not stabilised by the restriction, it is redistributed by it. A placebo that produces one positive in six on one sample and none on the other, together with an interaction term that crosses from an insignificant negative to a significant positive depending on which events are included, give a direct measure of how much apparent significance a design of this size generates on its own.

One caveat prevents over-reading the comparison at the longest horizon. A shock occurring after 2014 has no observed outcome at h = 5 within a panel ending in 2019, so at that horizon the restriction is not binding and the two samples coincide; the coefficients are accordingly identical. The restriction is informative from h = 0 to h = 3 and vacuous at h = 5.

4.3.3 *Discussion*

RQ2 receives a negative answer on its first half and no answer on its second. Affiliate presence does not rebound in the five years following a sudden disaster; if anything the point estimates drift downward, though only one is separately distinguishable from zero and that one is best read as a product of the number of tests performed. Whether the trajectory differs with historical exposure cannot be determined from estimates that hold no single sign across the window and are distinguishable from zero at no horizon within it.

On the first half, the result bears on the proposition reviewed in Section 2.4. The creative-destruction account was already contested at the level of aggregate output by Felbermayr & Gröschl (2014) and Hsiang & Jina (2014); Section 2.4.3 argued that this did not settle the question on the FDI margin, since reconstruction demand and modernised capital could raise inward investment while the national

income path remained depressed. On the present evidence that possibility is not realised for affiliate counts. The re-entry timing characterised by Kato & Okubo (2022) does not appear as a systematic return within five years, at least not at a level of aggregation where returns would have to be numerous enough to move country–sector cells.

Two comparisons follow. The rebound in FDI inflows three to five years after an event reported by Neise et al. (2022) is not reproduced here, and the caveat entered in Section 2.4.3 is the reason to be careful about calling this a contradiction: inflows are a flow and include reinvestment in affiliates that already exist, whereas affiliate counts are a stock. Capital can return to a host country without the number of affiliates changing, and on that reading the two findings are compatible. Against Oh & Oetzel (2011) the comparison is more direct and runs the other way: their null result at the affiliate level is corroborated here across a substantially broader set of host countries and industries, which addresses one of the two gaps identified in Section 2.6.

The second half of the question was motivated by the amplification result in Hsiang & Jina (2014), where cyclone losses are larger in countries with less historical exposure. That result concerns national income, and its analogue on affiliate counts is not detected here. Whether it is absent or merely undetectable is a question this design cannot settle: the interaction is identified from the intersection of shock timing with cross-country variation in exposure history, and the counts reported in Section 4.3.1 indicate how little of that intersection this panel contains.
There is also a reason internal to the outcome variable to expect a dated shock to be difficult to detect in it. Merlevede & Michel (2024) show that changes in the number of affiliates in European multinational networks arrive in restructuring episodes that frequently span several years, and that the largest networks adjust their portfolio of affiliates almost continuously. If the affiliate count moves principally in response to network strategy operating on a multi-year cadence, then a design anchoring the response to the calendar year of a disaster is looking for a signal in a series whose variation is dominated by something else. That is not a defect of the estimator but a statement about the outcome, and it belongs alongside the measurement issues taken up in Section 4.5.

Finally, the measurement of exposure bears on this result differently from the way it bears on RQ1. Sudden-onset events are the category EM-DAT records most completely, being discrete, visible and difficult to leave unreported, so RQ2 is estimated on the best-covered part of the database and the null is not readily explained by the hazard-type coverage bias documented in Section 3.1.1. What does bear on it is the spatial aggregation of the exposure measure, taken up in Section 4.5.2.

What the result supports can then be stated precisely. These estimates exclude a large and systematic rebound in affiliate presence in the five years following a sudden disaster, across a broad set of host countries and at the level of country–sector cells. They do not exclude a modest one, they say nothing about individual firms, and they do not establish that disasters leave multinational activity unaffected, only that no such effect is visible in this outcome at this level of aggregation.

4.4 GVC position and affiliate exit (RQ3)

RQ3 asks whether a host country-sector's position in the global value chain changes what a disaster does to affiliate presence, and whether any such moderation is itself conditioned by the host country's exposure history. Equation (1111) answers it through the triple interaction, the remaining terms being present because that interaction cannot be estimated without them. The estimates are reported first, followed by the robustness specifications and by a discussion of what a sample of this size allows a design of this kind to establish.

$$E[ENT_{ij,t+k} \mid \cdot] = exp(\alpha_{ij} + \lambda_t + \beta_1 \times D_{host,t} + \beta_2 \times \tilde{G}_{ij} + \beta_3 \times \tilde{H}_{host,t} + \beta_4 \times \left(D \times \tilde{G}\right)_{ij,t} + \beta_5 \times \left(D \times \tilde{H}\right)_{host,t} + \beta_6 \times \left(\tilde{G} \times \tilde{H}\right)_{ij,t} + \beta_7 \times \left(D \times \tilde{G} \times \tilde{H}\right)_{ij,t} + X\,\gamma) \quad (1111)$$

4.4.1 *Estimates*

One design choice must be stated before the estimates, because it departs from Section 4.3. There the shock is restricted to sudden-onset events; here it is measured across all disaster types. The reason is identifying variation, and it is a constraint rather than a preference. The value-chain position index is available for 56 host countries against the 185 in the panel, and sudden-onset events are the least frequent of the three hazard categories to begin with: 289 shock country-years across the full panel against 1,119 seasonal ones. Applied together, the two restrictions leave roughly 120 sudden-onset shock country-years at h = 0 and approximately 90 by h = 3, against the 409 and 299 that the pooled definition provides. A third-order interaction estimated on ninety events, spread across four horizons and 56 host countries, would return standard errors wide enough to accommodate almost any hypothesis, and the pooled definition is adopted in order to retain variation the design cannot afford to discard.

The cost of that choice is recorded here rather than argued away, and it takes two forms. The disaster term in this section is not the same object as the one in Section 4.3, so coefficients are not comparable across the two. And RQ3 has no falsification test corresponding to the seasonal placebo, because the events that would serve as the placebo are inside the treatment definition. Neither can be repaired within this sample. Both bear on how much weight the estimates below can carry, and Section 4.4.3 returns to them.

Table 4-5 reports seven coefficients on the disaster and moderator terms, of which one is the estimand. RQ3 asks whether the response to a shock depends jointly on chain position and exposure history, and that question is carried by the triple interaction alone. The remaining six are present because a triple interaction is not defined without its constituent parts, and because omitting any of them would load its variation onto the term of interest. With all continuous components apart from D mean-centred, each is read at the sample mean of the variables it is interacted with: the coefficient on the disaster term is the association with a shock for a country-sector of average chain position and average exposure history, not an average effect across the sample. None of the six is interpreted below except where stated.

On the estimand itself the answer is short. The triple interaction is +2.49 at impact, +7.39 at one year, +3.03 at two years and +1.24 at three, and only the second is distinguishable from zero. It is also the largest of the four by a factor of more than two, in a sequence that neither builds towards it nor falls away from it in any orderly fashion. The disaster term is insignificant at every horizon, and its interaction with chain position alone is insignificant throughout.

The arithmetic of the design supplies the context for that single result. Seven focal coefficients over four horizons, estimated on two outcome measures under two estimators, produce well over a hundred tests before the cohort restriction is added; at a five per cent threshold, five or six would be expected to cross it in the complete absence of any relationship. One isolated crossing, at a magnitude out of line with its neighbours, is what a design of this size generates on its own. Section 4.4.2 shows that it does not survive a change of estimator or of outcome measure, and that the discrete specification of the moderator diverges from it rather than confirming it.

Two of the remaining terms behave regularly enough within this table that passing over them in silence would invite the question. The interaction between chain position and historical exposure is positive and significant at all four horizons, between +0.77 and +0.94. Taken at face value it would say that upstream sectors are relatively more present in hosts with a longer disaster record, a

statement about levels rather than about a response to any event, and therefore not what RQ3 asks. The interaction between the disaster term and historical exposure is negative and marginally

Table 4-5: GVC Position and Disaster Shock Event Study (RQ3 - PPML)-Dependent Variable: Sectoral Affiliate Count (ENT)

Variables	T=0	T+1	T+2	T+3
Disaster Shocks & GVC Interactions				
Disaster severity shock (D_t)	+0.081	+0.077	+0.063	-0.049
	(0.475)	(0.544)	(0.652)	(0.647)
GVC position index ($G_{i,s}$, centered)	+0.156	+0.136	+0.415**	+0.525**
	(0.413)	(0.509)	(0.030)	(0.012)
Hist. disaster exposure ($H_{i,t}$, centered)	+0.133*	+0.051	+0.043	+0.057
	(0.074)	(0.487)	(0.549)	(0.435)
Disaster severity × GVC position ($D \times G$)	-0.217	+0.465	-0.154	-0.174
	(0.740)	(0.497)	(0.845)	(0.773)
Disaster severity × Hist. exposure ($D \times H$)	-0.859	-1.162*	-0.608	-1.073*
	(0.250)	(0.097)	(0.461)	(0.076)
GVC position × Hist. exposure ($G \times H$)	+0.938***	+0.767***	+0.817***	+0.797***
	(0.000)	(0.004)	(0.002)	(0.003)
Disaster × GVC × Hist. exposure ($D \times G \times H$)	+2.491	+7.394**	+3.034	+1.240
	(0.384)	(0.017)	(0.395)	(0.740)
Macroeconomic & Governance Controls				
Log GDP per capita (host, $t-1$)	+0.408**	+0.371**	+0.256	+0.016
	(0.011)	(0.020)	(0.228)	(0.844)
Log GDP per capita (home, $t-1$)	+0.840***	+0.737***	+0.692***	+0.396**
	(0.000)	(0.000)	(0.000)	(0.027)
Gov. effectiveness (host, $t-1$, centered)	-0.058	-0.047	-0.049	-0.007
	(0.130)	(0.214)	(0.270)	(0.861)
Model Specification & Statistics				
Dyad Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad
Observations	64,673	57,754	50,962	44,277
Dyad-years with disaster exposure	41,553	36,787	32,218	27,560
Distinct disaster country-years	409	372	337	299

*Notes: PPML estimates across Local Projections horizons $T = 0$ to $T + 3$ using Strategy B (NACE section level merged with OECD TiVA data). p-values in parentheses derived from standard errors clustered at the dyad level. *** $p < 0.01$,** $p < 0.05$, * $p < 0.1$. Continuous variables participating in interactions (G, H, Gov. Eff.) are mean-centered.*

significant at one and three years. Neither is part of the estimand, and Section 4.4.2 shows that the regularity of the first is confined to this specification. Finally, the variation available to identify the moderators belongs alongside the coefficients rather than after them. Chain position varies across country-sectors, and exposure history varies across host countries, of which there are 56. Standard errors are clustered at the dyad level and the number of dyads is large; the number of host countries carrying the variation in historical exposure is not. Inference on any term involving that moderator therefore rests on considerably less independent information than the observation count would suggest, a point taken up in Section 4.4.3.

4.4.2 *Robustness*

Comparison Table 4-6 answers the question that matters most directly. Estimated by log-OLS on the affiliate count, the triple interaction runs −0.46 at impact, +0.82 at one year and +2.96 at two, none of them distinguishable from zero. Whatever produced it, it is not a feature of the data that survives a change of estimator or of outcome measure.

The cohort restriction to shocks occurring between 2008 and 2016 tests something narrower than the two preceding checks, and it is worth being explicit about what it does and does not establish. Holding the set of identifying events fixed across horizons, the triple interaction runs +3.49 at impact, +7.86 at one year, +3.14 at two years and +1.24 at three, with only the second distinguishable from zero, the same pattern as the reported specification, and at that horizon a somewhat larger coefficient estimated more precisely. The result at one year is therefore not an artefact of which events enter at which horizon in this specification.

A third check replaces the continuous moderator with a discrete one. Splitting the sample at the median of host-country historical exposure and re-estimating with group-interaction dummies concentrates the test on the contrast between two exposure regimes, at the cost of the within-group variation it discards. The split does not corroborate the reported specification, it diverges from it. Significance appears on the subgroup-specific interaction between the disaster term and chain position among low-exposure hosts over the first three horizons, and not on the high-versus-low difference term that is the discrete analogue of the triple interaction. The pattern is estimator-dependent, and inference within each subgroup rests on roughly twenty-eight host countries. Appendix F reports the estimates in full.

Table 4-6: GVC Position Event Study Comparison — Enterprises (ENT) Log-OLS vs PPML 2008-2016 cohort

	Log-OLS ENT				PPML ENT 2008-2016			
Variables	**T=0**	**T+1**	**T+2**	**T=3**	**T+0**	**T+1**	**T+2**	**T+3**
Disaster severity shock (D_t)	+0.308***	+0.308***	+0.126	+0.033	+0.120	+0.071	+0.026	-0.049
	(0.006)	(0.009)	(0.266)	(0.803)	(0.295)	(0.579)	(0.861)	(0.647)
GVC position index ($G_{i,s}$, centered)	+0.081	+0.052	+0.105	+0.226	+0.059	+0.131	+0.433**	+0.525**
	(0.478)	(0.667)	(0.400)	(0.106)	(0.720)	(0.457)	(0.030)	(0.012)
Disaster severity × GVC position ($D \times G$)	-1.085*	-1.595***	-0.407	-0.448	-0.061	+0.769	+0.013	-0.174
	(0.068)	(0.006)	(0.453)	(0.598)	(0.940)	(0.313)	(0.988)	(0.773)
Disaster severity × Hist. exposure ($D \times H$)	-0.474	-0.921*	-0.764	-0.679	+0.090	-0.360	-0.651	-1.073*
	(0.339)	(0.066)	(0.154)	(0.219)	(0.849)	(0.438)	(0.451)	(0.076)
GVC position × Hist. exposure ($G \times H$)	+0.207	+0.268*	+0.239	+0.185	0.730***	+0.708***	+0.807***	+0.797***
	(0.164)	(0.092)	(0.154)	(0.320)	(0.002)	(0.006)	(0.001)	(0.003)
Disaster × GVC × Hist. exposure ($D \times G \times H$)	**-0.457**	**+0.822**	**+2.956**	**+4.955**	**+3.489**	**+7.863*** **	**+3.144**	**+1.240**
	(0.872)	(0.775)	(0.355)	(0.140)	(0.221)	(0.008)	(0.406)	(0.740)

*Notes: p-values in parentheses derived from standard errors clustered at the home-host-industry dyad level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Continuous variables participating in interactions (G, H) and governance controls are mean-centered.*

4.4.3 *Discussion*

RQ3 is not answered. The statement is worth making plainly, because everything that follows is an account of why, and because a question left open with reasons given is more useful than one closed on a single coefficient.
The question is not idle, and the failure to answer it should not be read as evidence that there is nothing to answer. Altay & Ramirez (2010) find position-dependent effects of flooding on firm asset turnover, with upstream and downstream firms moving in opposite directions, on firm-level data for multiple countries. That a design operating on country–sector cells across 56 hosts recovers no analogue is a statement about what this level of observation can resolve, not evidence that the conditioning they document is absent at the level where it was found.

The first reason is arithmetic. The estimand is a third-order interaction, which asks the data to detect a difference between differences between differences. It is estimated over four horizons on a moderator that varies across 56 host countries, and the median split leaves roughly 28 in each subgroup. Inference clustered at the dyad level counts a large number of dyads, but the variation identifying historical exposure originates at the host-country level, so the effective number of independent units is closer to the smaller figure. Cluster-robust standard errors rest on asymptotics in the number of clusters, and twenty-eight sits well below the range in which that approximation is ordinarily considered dependable (Cameron, C., & Miller, D., 2015).

The second is the design choice recorded at the start of this section. Pooling all hazard types preserved identifying variation, but it did so by admitting events whose timing is comparatively anticipable, which weakens the case that the estimates capture a response to a surprise rather than a correlation with severity. The section also has no falsification test, since the events that would supply one are inside the treatment definition. Both costs were accepted in exchange for a sample large enough to estimate the interaction at all, and neither is discharged by that exchange.

The third reason concerns which countries are in the sample, and it is the most consequential of the four. The chain-position index is constructed from input–output tables, which exist for economies with the statistical infrastructure to produce them. The 56 hosts covered are accordingly concentrated among high-income countries, and the 129 excluded are disproportionately those with the highest exposure to natural hazards and the least capacity to absorb them. The sample is therefore selected against precisely the variation the question requires. This is not a limitation that a longer panel or a different estimator would remedy, and it bounds what any version of this design can establish for as long as the moderator is measured in this way.

A fourth consideration follows from how that moderator is measured within the countries that are covered. The index is observed at NACE section level, and Section 3.1.4 and Appendix A showed, with the example of section B, that a single letter can span activities occupying opposite positions in the chain across two host countries. Measurement error in a moderator attenuates any interaction in which it participates, so a genuine moderating relationship would be estimated towards zero by construction. The null reported here is consistent with such a relationship existing and being unrecoverable at this level of sectoral aggregation.

What would answer RQ3 is therefore not a further specification of this design but a different one. Three routes are available in principle. Firm-level or subnational data would remove the aggregation affecting both the outcome and the exposure measure at once. A substantially larger set of host countries would supply the cross-country variation the moderator requires, at the cost of a chain-position measure constructed on a different basis. And a single-sector design followed in depth would trade breadth for the ability to observe position directly rather than through a sectional average.

Chapter 5 takes these up. What can be said on the present data is that the question is well posed and that this sample cannot settle it.

4.5 What the data can and cannot identify

Each of the three preceding sections ended by referring a difficulty to this one, and the difficulties they referred are properties of the data rather than of any particular estimator. They are gathered here for that reason, rather than repeated three times. What follows sets out how exposure is measured, how the two aggregations built into the outcome bear on what can be observed through it, and how much identifying variation the design actually contains. The purpose is not to qualify the results but to establish what they are results about.

4.5.1 The measurement of exposure

EM-DAT is a reporting system before it is a hazard register. An event enters the database when it crosses one of four humanitarian thresholds and when someone records that it has done so, and Delforge et al. (2025) identify four systematic consequences: reporting has increased over time, its completeness varies across countries, slow-onset hazards are under-represented relative to sudden ones, and the inclusion thresholds censor the lower tail of severity. Section 3.1.1 documented the second of these directly on the countries used here, and found what the concern implies: reported frequency per capita rises with government effectiveness, and median per-event severity falls with it. Countries with weaker institutions do not record fewer disasters because fewer occur; they record the extreme ones and omit the moderate ones.

What makes this more than a general caution is that the bias runs along the same axis as one of the variables the design uses. Government effectiveness is the moderator in RQ1 and a control in every panel specification, and it covaries with the measurement properties of the exposure variable it is set against. The exposure measure is therefore not independent of the terms it is interacted with, and the consequence is not the same across the three questions.

For RQ1 it is direct. A positive coefficient on reported historical frequency is consistent with exposed places being attractive, and equally consistent with reporting capacity being associated with affiliate presence for reasons unconnected to hazard. The two readings are not distinguished by anything in the cross-section, and the negative interaction is what the second of them predicts: where reporting is sparse a recorded count carries information about much besides hazard, and where reporting is complete it carries less, so the marginal association should weaken as governance rises. Section 4.2.1 reports that it weakens to approximately zero at the top of the observed governance range.

For RQ2 the bias runs the other way and is worth stating for that reason. Sudden-onset events are the category the database records most completely, being discrete, visible and difficult to leave unrecorded, so the design is estimated on the best-covered part of the data. The absence of a rebound is not readily attributed to a hazard that went unrecorded.

For RQ3 the position is less favourable. Pooling all hazard types, for the reasons given in Section 4.4, means inheriting the hazard-type bias in full, including the under-representation of slow-onset events whose effects on affiliate retention would in any case be the hardest to date. The threshold bias applies throughout and bounds what any redesign within this source could achieve: because the lower tail of severity is censored by construction, a measure built on severity rather than counts still varies only over the part of the distribution EM-DAT already considers severe. Separating hazard from reporting requires a measure constructed independently of the reporting system, and Chapter 5 takes up what that would involve.

4.5.2 Spatial aggregation

Exposure is observed at the country-year level while affiliates are located at points within countries, and the two do not correspond. A flood affecting one province of China enters the panel as a shock to every Chinese cell, including affiliates two thousand kilometres from the water; the same event in Tuvalu would affect essentially every establishment in the country. The variable therefore measures something close to what firms experience in small hosts and something considerably weaker in large ones.

The resulting error attenuates estimates towards zero, which would be unremarkable if it were classical. It is not. The mismatch between the measured shock and the shock any given affiliate experiences grows with the geographic extent of the host country, and the largest hosts are also those holding the largest number of affiliates. Attenuation is accordingly strongest in precisely the cells that contribute most of the variation in the outcome, so a true response of moderate size would be expected to appear as a null in aggregate even if it were present in every affected region.

This bears unevenly on the three questions, and the pattern is informative. RQ1 uses a long-run frequency measure that aggregates over many events in many locations, so a country-level average is a reasonable summary of what a firm located anywhere in that country faces over a decade. RQ2 and RQ3 date a single event, for which no such averaging applies: the country-level indicator is either approximately right for a given affiliate or almost entirely wrong, and which of the two it is cannot be determined from the data. That the cross-sectional question yields a stable result while the two event-study questions do not is consistent with this difference before any economic mechanism is invoked.

4.5.3 Cell aggregation

The outcome is aggregated in a second way, and this one is a property of the source rather than of the merge. FATS reports counts and employment totals for home–host–industry–year cells; no affiliate is identified within a cell and none can be followed across years. A cell recording five affiliates and two thousand employees is consistent with one establishment of 1,996 employees alongside four of one, and equally consistent with five of four hundred, and the two are indistinguishable in these data. Three consequences follow, all of them visible in the results above.

First, affiliate counts and employment are not two measures of one quantity, so their divergence across specifications in Sections 4.3.2 and 4.4.2 is not necessarily a contradiction. A network that consolidates activity into fewer, larger establishments moves the two in opposite directions, and nothing in these data separates that from a change in total activity.

Second, exit is observed only in net terms. A cell falling from forty-five affiliates to forty records that five more affiliates left than entered, which is genuine information about exit; what it does not record is how many left, how many entered, or which ones did either. Gross flows are invisible, no affiliate can be followed from one year to the next, and no departure can be attributed to an affiliate that experienced the disaster rather than to one elsewhere in the same country and industry.

Third, and least visible in the tables, the confidentiality rule interacts with the first two. Eurostat suppresses cells containing one or two affiliates, which produces missing rates of 12.9 per cent for the affiliate count, 20.5 per cent for employment and 21.8 per cent for turnover. The suppressed cells are the smallest ones: precisely those in which the departure of a single affiliate would move the recorded count furthest, and in which an exit is therefore most detectable. The censoring is not independent of the quantity the event studies are designed to observe.

Merlevede & Michel (2024) make the same distinction from the other side, noting that a dataset identifying complete multinational networks at the level of individual affiliates differs in kind from the foreign affiliates statistics collected by most European countries, which are organised around country-pair aggregates. The present design uses the second, and inherits its structure along with its coverage.

These three features bear on the results in a way the chapter has already illustrated rather than needing to restate. The estimates that moved most between specifications — the RQ2 interaction that crossed from an insignificant negative to a significant positive when the shock cohort was restricted, and the RQ3 term whose significance shifted from four horizons to one when the outcome measure changed — are the estimates drawn from the smallest amounts of identifying variation and subject to the coarsest aggregation. The cross-sectional estimate of Section 4.2, which uses every pair in the sample and aggregates exposure over two decades, does not move at all. What separates the stable result from the unstable ones in this chapter is not the plausibility of the mechanisms behind them, but the quantity and the resolution of the data available to test each.

4.6 Summary of findings

Table 4-7: What the estimates establish for each research question, and what bounds it

Question	What the estimates show	Answer	What limits it
RQ1	A positive association between historical exposure and affiliate presence, attenuated as governance rises but not reversed anywhere within the observed range	positive in sign	Exposure measured through a reporting system that covaries with the moderator; cross-section with no exogenous variation
RQ2	A negative trajectory with no reversal at any horizon; the moderation term is negative over the first three horizons and positive over the last three, and significant at none	No rebound; the moderation question unanswered	Imprecision at the longest horizon; country-level measurement of a sub-national event
RQ3	One significant coefficient across four horizons, absent under the alternative estimator and the alternative outcome measure, and not corroborated by the discrete specification of the moderator	Unanswered	56 host countries; value-chain coverage concentrated among high-income economies; moderator observed at section level

5 CONCLUSION

5.1 Main Findings

This thesis asked three questions about the relationship between natural disasters and the presence of European multinational affiliates abroad, using Eurostat Foreign Affiliates Statistics for 29 home countries and up to 185 hosts over 2008–2019, matched to exposure measures constructed from EM-DAT. None of the three is answered with confidence, and the reasons differ across them.

The first question produces the clearest pattern in the data and the least secure interpretation of it. Host countries with a longer record of reported disasters are associated with a greater presence of European affiliates, and the association is stable across every variation applied to it: a different outcome measure, a different margin of presence and a different functional form all leave its sign and its order of magnitude intact. Institutional quality attenuates it without reversing it anywhere within the range of governance the sample contains. What the estimates cannot establish is what the association holds between. The exposure variable counts events that were reported, and Section 3.1.1 showed that reported frequency covaries with government effectiveness while median per-event severity moves against it. A positive coefficient is accordingly consistent with exposed places being attractive for reasons of geography, and equally consistent with reporting capacity being associated with affiliate presence for reasons unconnected to hazard. Nothing in a cross-section of this kind separates the two, and Section 4.2 sets out why the sign of the interaction does not separate them either.

The second question receives a negative answer on its first half and none on its second. Affiliate presence does not rebound in the five years following a sudden-onset disaster: the estimated path is negative at every horizon, so the trajectory that the creative-destruction proposition reviewed in Section 2.4 would imply fails on its sign pattern before the question of statistical significance arises. Whether that path differs with the host country's exposure history cannot be determined from these estimates, since the relevant coefficient holds no single sign across the window and is distinguishable from zero at no horizon within it.

The third question is not answered at all. One significant coefficient appears among the estimates, at a single horizon, and it is reproduced neither under an alternative estimator, nor on an alternative outcome measure.

5.2 What the design rules out

That the estimates settle so little raises an obvious objection, and the sequence of specifications reported in Chapters 3 and 4 was assembled to answer it. Each of the ordinary explanations for a result of this kind was addressed by a specific choice, and each can accordingly be set aside.

Multiplicative heteroskedasticity, and the bias it introduces into log-linear estimation, is handled by Poisson pseudo-maximum-likelihood, which also retains the zero-valued cells that log-linear estimation discards, the observations most informative about complete exit. Time-invariant heterogeneity between country pairs is absorbed by dyad fixed effects, common macroeconomic shocks by year fixed effects. Collinearity among the interaction terms is reduced by mean-centring and checked by variance inflation factors, serial correlation within dyads by clustering at that level, and compositional change across horizons by restricting the shock cohort so that the same events identify every horizon. The possibility that an isolated significant coefficient is a product of the number

of tests performed is addressed by the seasonal placebo and by a stated convention against interpreting any coefficient on the strength of its own p-value.

What none of these choices can address is the data themselves, and that is the residual. The exposure measure records what was reported rather than what occurred; it is observed at country resolution while affiliates occupy points within countries; and the outcome is aggregated into cells within which no affiliate can be identified. What this thesis reports is therefore in large part a set of statements about what these sources can and cannot identify.

5.3 Contribution

Stated as findings rather than as shortcomings, three propositions follow, and together with the estimates they constitute what this thesis contributes.

The first travels furthest beyond the present exercise. The reporting properties of EM-DAT are not a background caveat but covary with the institutional variables this literature uses as moderators. Section 3.1.1 documents that covariance directly, and Section 4.2 shows it operating inside the estimates. An estimate of moderation by governance is therefore not separable from an estimate of moderation by reporting capacity, absent an exposure measure constructed independently of the reporting system. This bears on any study interacting an EM-DAT-derived exposure measure with an index of institutional quality, which describes a substantial share of the empirical literature reviewed in Chapter 2, and it bears on the results reported here as much as on any other.

The second concerns dated responses. Questions attaching a response to a specific event are not answerable with these sources, and the reason is structural rather than incidental: a country-year exposure indicator does not identify which affiliates were exposed, and a cell-aggregated outcome does not identify which affiliates left. The absence of a rebound reported here is consistent with the affiliate-level null of Oh & Oetzel (2011), obtained across a substantially broader set of host countries and industries than that study covered. It is not inconsistent with the rebound in FDI inflows reported by Neise et al. (2022) either: inflows are a flow that includes reinvestment in affiliates that already exist, affiliate counts are a stock, and capital can return to a host country without the number of affiliates changing.

The third concerns the difference between precision and interpretability. The cross-sectional estimates are precise and stable while the event-study estimates are neither, and it would be natural to read the first as the more secure of the two. The first proposition above is the reason that reading is unavailable: an estimate can be robust to every variation applied to it and remain uninterpretable, if what varies across specifications is not the source of the ambiguity. The contrast between the two halves of Chapter 4 illustrates that distinction rather than resolving it.

One methodological point supports the comparisons drawn above. The three questions are estimated on a single dataset, under the same fixed effects, the same estimator hierarchy and the same reporting conventions, which is what allows the stability of the cross-sectional estimates and the instability of the event-study estimates to be read against each other rather than as separate exercises. They are not, however, estimated jointly: each question varies one moderator while holding the others outside the specification, and RQ3 pools hazard types for the reason given in Section 4.4. Whether the three survive conditioning on one another remains open, and Section 5.5 returns to it.

5.4 Limitations

Section 4.5 sets out the three properties of the data that bound what any of these estimates can establish, and they are not restated here. Three further limitations belong to the design rather than to the sources.

The empirical strategy is associational throughout. Nothing in it supplies exogenous variation in exposure, and no causal claim is made or implied at any point.

The hazard classification separates families of events rather than events. Sudden-onset events are treated as approximately unexpected and seasonal ones as approximately expected, an approximation that fails in both directions and, as Section 3.1.1 records, fails asymmetrically.

Finally, the panel window coincides with a structural change in the outcome variable that has nothing to do with natural hazards. Merlevede & Michel (2024) document a shift in European multinational networks over precisely this period, with foreign contraction episodes displacing expansions in the second half of the decade. Year fixed effects absorb the common component, but affiliate counts moved over these years for reasons of network strategy whose scale bears no comparison with a response to a natural hazard.

5.5 Future Work

Four extensions follow from the diagnosis above. They are distinguished by what each would repair and at what cost, rather than offered as a programme.

The narrowest sharpens the falsification test without replacing any data. Classifying the events themselves — establishing, for each seasonal event, whether it fell inside or outside the recurrent pattern of its region — would produce a placebo group that is expected in fact rather than by category, and would allow the anticipation channel to be tested rather than assumed. The work is feasible on a database of this size but is manual, and it becomes more necessary over time, since a warming climate is moving hazards into the expected distribution in regions that had not previously faced them.

A second changes the exposure measure within the same source. Historical exposure is measured here by event counts; the severity-based alternative was set aside for the reason given in Section 3.1.1, namely that severity aggregated over two decades is dominated by whichever single event in the window was largest, which is the quantity the threshold and geography biases distort most. Whether a severity-weighted measure nonetheless carries information the count does not is a question that could be settled directly. The related suggestion of excluding moderate events to make the sample more even was considered and set aside: the inclusion threshold already censors the lower tail of severity, so raising it further would deepen the bias documented in Section 3.1.1 rather than correct it, and the cut point would be arbitrary.

A third changes the outcome. The multinational network panel constructed by Merlevede and Theodorakopoulos (2024) identifies affiliates individually, so that exit becomes the exit of an entity rather than the disappearance of a cell. One constraint should be stated: that source records location and ownership for affiliates worldwide but activity data only for those located in Europe, and most disaster-exposed hosts are not. The extension would deliver exit without scale.

The fourth changes both, and is the one this diagnosis points to most directly. Bijnens et al. (2024) identify exposure from rainfall records and firm geolocation rather than from a disaster database, and

follow affected firms and their trading partners through transaction-level data. That construction addresses all three of the properties set out in Section 4.5 at once. Its cost is coverage — one event in one country, against every recorded event across 185 — and the trade-off is the exact inverse of the one made here.

The three questions posed in this thesis remain well posed. What it establishes is narrower than what it set out to obtain: that the sources on which this literature commonly relies record disaster exposure in a way that is neither independent of the institutions used to moderate it, nor resolved finely enough to attach a response to a particular event, and that both properties bear on the estimates reported here as much as on those reported elsewhere. Establishing where the limits of a measurement apparatus lie is a smaller contribution than answering the questions the apparatus was assembled to answer, and a more durable one.

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APPENDICES

Appendix A Industry Classification and Harmonisation

Industry activity in the panel is recorded using the NACE Rev.2 classification at the section level (letters A–U). Two harmonisation issues arise before this classification can be used consistently across the full 2008–2019 panel: (i) reconciling NACE Rev.2 with ISIC Rev.4, the classification used by the OECD TiVA database, and (ii) reconciling NACE Rev.2 with NACE Rev.1.1, which Eurostat's FATS database still used for the 2008–2009 reporting years. Appendix - Table i presents the reference nomenclature; Appendix - Table ii documents the Rev.1.1 → Rev.2 crosswalk applied to the early panel years.

At the section level, NACE Rev.2 and ISIC Rev.4 use identical codes for identical activities, so no crosswalk is required between these two classifications — Appendix - Table i therefore serves as the common reference for both FATS (NACE Rev.2) and TiVA (ISIC Rev.4) data. Sections T and U (household and extraterritorial activities) are not covered by TiVA and are consequently excluded from any specification involving GVC variables.

Appendix - Table i: NACE Rev.2 / ISIC Rev.4 Section Codes

Code	Description	ISIC Rev.4 Divisions	In FATS	In TiVA
A	Agriculture, forestry and fishing	01–03	Yes	Yes
B	Mining and quarrying	05–09	Yes	Yes
C	Manufacturing	10–33	Yes	Yes
D	Electricity, gas, steam and air conditioning supply	35	Yes*	Yes
E	Water supply; sewerage, waste management and remediation	36–39	Yes*	Yes
F	Construction	41–43	Yes	Yes
G	Wholesale and retail trade; repair of motor vehicles	45–47	Yes	Yes
H	Transportation and storage	49–53	Yes	Yes
I	Accommodation and food service activities	55–56	Yes	Yes
J	Information and communication	58–63	Yes	Yes
K	Financial and insurance activities	64–66	Yes	Yes
L	Real estate activities	68	Yes	Yes
M	Professional, scientific and technical activities	69–75	Yes	Yes
N	Administrative and support service activities	77–82	Yes	Yes
O	Public administration and defence; compulsory social security	84	Yes	Yes
P	Education	85	Yes	Yes
Q	Human health and social work activities	86–88	Yes	Yes
R	Arts, entertainment and recreation	90–93	Yes	Yes
S	Other service activities	94–96	Yes	Yes
T	Activities of households as employers	97–98	Yes	No
U	Activities of extraterritorial organizations and bodies	99	Yes	No

** Sections D and E are only separately identifiable in FATS from 2010 onward. For 2008–2009, source data were still reported under NACE Rev.1.1, whose section E (Electricity, gas and water supply) splits ambiguously across Rev.2 D and E (see Table X.2); affected rows are dropped for those two years.*

For 2008–2009, FATS reports activity under the legacy NACE Rev.1.1 classification. Each Rev.1.1 code was mapped to its Rev.2 equivalent; codes with no unambiguous one-to-one mapping were dropped for these two years rather than assigned arbitrarily, producing a known, documented coverage gap in the early panel.

Appendix - Table ii: NACE Rev.1.1 → Rev.2 Crosswalk (FATS 2008–2009 only)

Rev.1.1 Code	Rev.1.1 Description	Mapped to (Rev.2)	Mapping Type	Note
A	Agriculture and hunting	A	Direct	Merged with old B under Rev.2 A
B	Fishing	A	Merged	Absorbed into Rev.2 A
A_B	Agriculture + fishing (combined)	A	Direct	
C	Mining and quarrying	B	Direct	
D	Manufacturing	C	Direct	
F	Construction	F	Direct	
G	Wholesale and retail trade	G	Direct	
H	Hotels and restaurants	I	Direct	Renamed: Accommodation & food service
J	Financial intermediation	K	Direct	Renamed: Financial & insurance activities
L	Public administration	O	Direct	
M	Education	P	Direct	
N	Health and social work	Q	Direct	
P	Private households	T	Direct	
Q	International organizations	U	Direct	
C-O_X_L	Total, excl. public administration (agg.)	B-S_X_O	Aggregate	
G-O_X_L	Services, excl. public administration (agg.)	G-S_X_O	Aggregate	
A-O_X_L	Total, excl. public administration (agg.)	—	Dropped	No clean Rev.2 aggregate equivalent
E	Electricity, gas and water supply	D + E	Dropped	Splits into two Rev.2 sections; no unambiguous assignment
I	Transport, storage and communication	H + J	Dropped	Splits into Transport (H) and ICT (J)
K	Real estate, renting and business activities	L + M + N	Dropped	Splits into Real estate, Professional, and Admin support
O	Other community, social and personal services	R + S	Dropped	Splits into Arts (R) and Other services (S)

Four Rev.1.1 sections (E, I, K, O) split into multiple Rev.2 sections with no basis for an unambiguous one-to-one assignment; observations carrying these codes are excluded from the panel for 2008–2009 only. This affects the electricity/water, transport/ICT, real estate/professional services, and arts/other services sectors specifically in the first two panel years, and is reported here as a limitation rather than resolved through an arbitrary allocation rule.

Section B illustrates why a single section letter can span very different chain positions. In a resource economy such as the DRC, section B is dominated by extraction of metal ores for export with minimal domestic processing, placing the sector firmly upstream (Cobalt Institute, 2022). Belgium, by contrast, has no coal, oil and gas, or metal-ore extraction of note, the corresponding OECD sub-activity codes being zero throughout the sample. Its section B activity is almost entirely construction-material quarrying (limestone, sand, gravel) feeding a substantial domestic cement and construction-materials industry (Loyd M., 2025). Because these commodities are mostly consumed by downstream users within the same country rather than exported over long distances, Belgium's section B position hovers close to neutral rather than taking the strongly upstream values seen in resource-extraction economies.

Appendix B PPML

Appendix B.1 Estimation via IRLS

PPML assumes an exponential conditional mean,

$$E[y_i|X_i] = \mu_i = e^{X_i\beta} \tag{12}$$

and is estimated by pseudo-maximum likelihood. Because the Poisson log-likelihood belongs to the linear exponential family, the estimator is consistent whenever (12) is correctly specified, irrespective of the true distribution of y_i(Santos Silva & Tenreyro, 2006). Estimation proceeds by iteratively reweighted least squares: at each iteration the model is linearised around the current fit, yielding a working variable

$$z_i^{(m)} = X_i\beta^{(m)} + \frac{\left(y_i - \mu_i^{(m)}\right)}{\mu_i^{(m)}} \tag{13}$$

with weights $\omega_i^{(m)} = \mu_i^{(m)}$, and β is updated by weighted least squares. Iteration continues until $\| \beta^{(m+1)} - \beta^{(m)} \| < 10^{-6}$.

Appendix B.2 Fixed-effects absorption

As with the linear TWFE model, explicit dummies for every dyad and year are infeasible at this scale. Within each IRLS iteration, pyfixest's fepois implementation applies the alternating-projections algorithm of Gaure (2010) to the working variable $z_i^{(m)}$:the dyad-weighted mean is swept out, then the year-weighted mean, and the two steps are repeated until the within-group variance is fully eliminated, before the weighted least-squares update is applied.

Appendix C VIF

Appendix - Table iii: VIF RQ1 — Model 3 regressors (main spec)

Variable	VIF
ln GDPpc_host	4.6882
ln Pop_host	4.2735
ln GDPpc_home	1.0920
ln Pop_home	1.0863
ln Dist	1.5993
Contig	1.2687
Colony	1.5998
Comlang	1.6042
Landlocked_host	1.1375
ln Area_host	3.1275
GE_host	4.5279
ln HistFreq_host	2.0862
GE_host x HistFreq_host	1.1443

Appendix D RQ1: Robustness checks

Appendix - Table iv: Robustness I — ln_avg_emp (2008–2019 average)

Dep. var: ln_avg_emp (2008–19 average)	R1-M1	R1-M2	R1-M3
Variable			
Constant	-49.1197***	-50.4044***	-51.4036***
	(0.0000)	(0.0000)	(0.0000)
ln GDPpc_host	0.5773***	0.6052***	0.6665***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_host	0.8161***	0.8985***	0.9160***
	(0.0000)	(0.0000)	(0.0000)
ln GDPpc_home	2.1462***	2.1617***	2.1668***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_home	1.2885***	1.2919***	1.2982***
	(0.0000)	(0.0000)	(0.0000)
ln Dist	-0.9603***	-1.0147***	-1.0183***
	(0.0000)	(0.0000)	(0.0000)
Contig	1.3718***	1.2911***	1.2635***
	(0.0000)	(0.0000)	(0.0000)
Colony	1.6705***	1.6859***	1.6601***
	(0.0000)	(0.0000)	(0.0000)
Comlang	0.8068***	0.7733***	0.7905***
	(0.0006)	(0.0010)	(0.0007)
Landlocked_host	-0.4329***	-0.4313***	-0.3775***
	(0.0018)	(0.0019)	(0.0068)
ln Area_host	0.0341	0.0280	0.0310
	(0.2976)	(0.3936)	(0.3416)
GE_host	0.3812***	0.3441***	0.2573**
	(0.0005)	(0.0015)	(0.0242)
ln HistFreq_host		0.2476***	0.3109***
		(0.0013)	(0.0003)
GE_host x HistFreq_host			-0.2310***
			(0.0053)
N	2,337	2,337	2,337
Adj. R^2	0.5804	0.5819	0.5836

Appendix - Table v: Robustness II — ever_active, LPM

Dep. var: ever_active (LPM)	LPM-M1	LPM-M2	LPM-M3
Variable			
Constant	-4.0905***	-4.2316***	-4.2665***
	(0.0000)	(0.0000)	(0.0000)
ln GDPpc_host	0.0616***	0.0687***	0.0714***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_host	0.0651***	0.0748***	0.0754***
	(0.0000)	(0.0000)	(0.0000)
ln GDPpc_home	0.2499***	0.2504***	0.2503***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_home	0.0888***	0.0888***	0.0888***
	(0.0000)	(0.0000)	(0.0000)
ln Dist	-0.1101***	-0.1189***	-0.1192***
	(0.0000)	(0.0000)	(0.0000)
Contig	0.0434	0.0251	0.0212
	(0.2648)	(0.5220)	(0.5906)
Colony	0.2489***	0.2512***	0.2509***
	(0.0000)	(0.0000)	(0.0000)
Comlang	-0.1270***	-0.1307***	-0.1295***
	(0.0000)	(0.0000)	(0.0000)
Landlocked_host	-0.0249*	-0.0198	-0.0173
	(0.0858)	(0.1699)	(0.2353)
ln Area_host	-0.0008	-0.0009	-0.0006
	(0.8352)	(0.8006)	(0.8676)
GE_host	0.0629***	0.0542***	0.0493***
	(0.0000)	(0.0000)	(0.0000)
ln HistFreq_host		0.0301***	0.0334***
		(0.0000)	(0.0000)
GE_host x HistFreq_host			-0.0117*
			(0.0619)
N	5,481	5,481	5,481
Adj. R^2	0.4063	0.4084	0.4087

Appendix - Table vi: Robustness III — ever_active, Logit AME

Dep. var: ever_active (Logit AME)	Logit-M1 AME	Logit-M2 AME	Logit-M3 AME
Variable			
ln GDPpc_host	0.0535***	0.0623***	0.0648***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_host	0.0633***	0.0767***	0.0777***
	(0.0000)	(0.0000)	(0.0000)
ln GDPpc_home	0.2332***	0.2342***	0.2345***
	(0.0000)	(0.0000)	(0.0000)
ln Pop_home	0.0828***	0.0829***	0.0829***
	(0.0000)	(0.0000)	(0.0000)
ln Dist	-0.1085***	-0.1209***	-0.1217***
	(0.0000)	(0.0000)	(0.0000)
Contig	0.2479***	0.2198***	0.2175***
	(0.0015)	(0.0046)	(0.0050)
Colony	0.2698***	0.2694***	0.2663***
	(0.0000)	(0.0000)	(0.0000)
Comlang	-0.1351***	-0.1367***	-0.1356***
	(0.0000)	(0.0000)	(0.0000)
Landlocked_host	-0.0127	-0.0050	-0.0035
	(0.3430)	(0.7127)	(0.7931)
ln Area_host	0.0018	0.0013	0.0014
	(0.6073)	(0.7068)	(0.6777)
GE_host	0.0701***	0.0600***	0.0554***
	(0.0000)	(0.0000)	(0.0000)
ln HistFreq_host		0.0392***	0.0436***
		(0.0000)	(0.0000)
GE_host x HistFreq_host			-0.0098
			(0.1383)

Appendix E RQ2: Robustness checks

Appendix - Table vii: Robustness I ENT OLS

Variables	T=0	T+1	T+2	T+3	T+4	T+5
Disaster Shocks & Interactions						
Sudden physical shock	+0.3528	+0.1702	-0.3071	-0.5231	-0.2245	-0.668*
	(0.301)	(0.650)	(0.390)	(0.404)	(0.598)	(0.095)
Sudden shock × Hist. disaster freq.	-0.1746	-0.2962	-1.1317	-0.4686	+0.2986	+0.0492
	(0.890)	(0.776)	(0.180)	(0.713)	(0.823)	(0.957)
Seasonal physical shock (placebo)	-0.0332	+0.0442	+0.0883	+0.0640	-0.1129	-0.0365
	(0.768)	(0.686)	(0.521)	(0.636)	(0.487)	(0.808)
Macroeconomic & Governance Controls						
Log GDP per capita (host, $t-1$)	+0.5125***	+0.4550***	+0.3700***	+0.2286***	+0.1724**	+0.0771
	(0.000)	(0.000)	(0.000)	(0.005)	(0.045)	(0.378)
Log GDP per capita (home, $t-1$)	+0.9747***	+0.8723***	+0.6063***	+0.4493***	+0.2286	+0.0733
	(0.000)	(0.000)	(0.000)	(0.002)	(0.134)	(0.645)
Gov. effectiveness (host, $t-1$, centered)	-0.0149	+0.0057	+0.0333	+0.0759**	+0.0615*	+0.0449
	(0.649)	(0.866)	(0.325)	(0.030)	(0.085)	(0.352)
Hist. disaster frequency (host, dynamic)	-0.0466	-0.0499	-0.0248	+0.0083	+0.0256	+0.0489
	(0.347)	(0.325)	(0.640)	(0.875)	(0.628)	(0.357)
Model Specification & Statistics						
Dyad Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad	Dyad	Dyad
Observations	21,744	18,983	16,680	14,581	12,573	10,652
Distinct sudden disaster country-years	281	248	226	199	174	155
Distinct seasonal disaster country-years	1,093	983	890	793	699	607
Adjusted R^2	0.028	0.022	0.013	0.007	0.003	0.001

*Notes: Log-OLS estimates across Local Projections horizons $T=0$ to $T+5$. p-values in parentheses derived from standard errors clustered at the home-host-industry dyad level. *** $p<0.01$, $p<0.05$, * $p<0.1$. Continuous interaction terms are mean-centered.*

Appendix - Table viii: Robustness II EMP OLS

Variables	T=0	T+1	T+2	T+3	T+4	T+5
Disaster Shocks & Interactions						
Sudden physical shock	-0.9897	-0.6728	-0.4691	-1.7330**	+0.1979	+0.7034
	(0.352)	(0.547)	(0.476)	(0.036)	(0.771)	(0.287)
Sudden shock × Hist. disaster freq.	+2.3762	+1.4284	-1.7099	+1.9103	-1.5784	-1.7649
	(0.407)	(0.709)	(0.485)	(0.515)	(0.529)	(0.394)
Seasonal physical shock (placebo)	-0.2360	-0.0312	-0.1476	+0.2292	+0.1625	-0.0235
	(0.494)	(0.900)	(0.537)	(0.435)	(0.554)	(0.937)
Macroeconomic & Governance Controls						
Log GDP per capita (host, $t-1$)	+0.8848***	+0.6007***	+0.3399**	+0.0393	-0.0576	-0.1769
	(0.000)	(0.000)	(0.036)	(0.823)	(0.746)	(0.310)
Log GDP per capita (home, $t-1$)	+2.0003***	+1.8250***	+1.7075***	+1.6340***	+1.6772***	+1.3292***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Gov. effectiveness (host, $t-1$, centered)	-0.1060	-0.0857	-0.0361	-0.0299	-0.0658	-0.0931
	(0.153)	(0.210)	(0.580)	(0.656)	(0.356)	(0.336)
Hist. disaster frequency (host, dynamic)	+0.0495	+0.0744	+0.0807	+0.1568	+0.2359*	+0.2191*
	(0.675)	(0.547)	(0.507)	(0.179)	(0.096)	(0.064)
Model Specification & Statistics						
Dyad Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad	Dyad	Dyad
Observations	15,755	13,363	11,698	10,177	8,756	7,437
Distinct sudden disaster country-years	258	229	205	181	162	140
Distinct seasonal disaster country-years	1,005	892	801	710	621	535
R^2	0.035	0.027	0.020	0.018	0.019	0.013

*Notes: Log-OLS estimates across Local Projections horizons $T = 0$ to $T + 5$. p-values in parentheses derived from standard errors clustered at the home-host-industry dyad level. *** $p < 0.01$, $p < 0.05$, * $p < 0.1$. Continuous interaction terms are mean-centered*

Appendix - Table ix: Robustness III PPML EMP

Variables	T=0	T+1	T+2	T+3	T+4	T+5
Disaster Shocks & Interactions						
Sudden physical shock	-0.610	+0.041	-0.368	-0.970*	-0.181	+0.618
	(0.324)	(0.929)	(0.496)	(0.068)	(0.561)	(0.368)
Sudden shock × Hist. disaster freq.	-1.799	-1.185	-1.121	+1.120	+0.336	+3.817***
	(0.325)	(0.548)	(0.556)	(0.485)	(0.830)	(0.002)
Seasonal physical shock (placebo)	-0.475	-0.021	-0.141	+0.121	+0.550**	-0.124
	(0.253)	(0.950)	(0.600)	(0.318)	(0.043)	(0.454)
Macroeconomic & Governance Controls						
Log GDP per capita (host, $t-1$)	+0.7549***	+0.5933***	+0.3778**	+0.1156	+0.1049	-0.1506
	(0.000)	(0.000)	(0.002)	(0.387)	(0.431)	(0.195)
Log GDP per capita (home, $t-1$)	+0.7581***	+0.7973***	+0.8830***	+0.8170***	+0.8439***	+0.8248***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Gov. effectiveness (host, $t-1$, centered)	-0.0362	-0.0074	+0.0154	+0.0392	-0.0124	+0.0200
	(0.457)	(0.888)	(0.760)	(0.401)	(0.782)	(0.727)
Hist. disaster frequency (host, dynamic)	+0.2053*	+0.1845*	+0.1752*	+0.1975**	+0.1490*	+0.0699
	(0.058)	(0.069)	(0.052)	(0.026)	(0.093)	(0.342)
Model Specification & Statistics						
Dyad Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad	Dyad	Dyad
Observations	17,223	15,482	13,806	12,112	10,494	8,924
Distinct sudden disaster country-years	286	256	234	210	188	168
Distinct seasonal disaster country-years	1,109	1,006	913	821	733	631
Pseudo R^2	0.986	0.987	0.988	0.990	0.991	0.993

*Notes: Log-OLS estimates across Local Projections horizons $T = 0$ to $T + 5$. p-values in parentheses derived from standard errors clustered at the home-host-industry dyad level. *** $p < 0.01$, $p < 0.05$, * $p < 0.1$. Continuous interaction terms are mean-centered.*

Appendix - Table x: Robustness IV: ENT 2008-2014 cohort

Variables	T=0	T+1	T+2	T+3	T+4	T+5
Panel A: Log-OLS Estimates						
Sudden physical shock	+0.175	-0.150	-0.244	-0.102	+0.102	-0.668*
	(0.662)	(0.664)	(0.594)	(0.858)	(0.836)	(0.095)
Sudden shock × Hist. disaster freq.	+2.531**	+0.797	-0.605	-0.676	-0.143	+0.049
	(0.021)	(0.387)	(0.453)	(0.543)	(0.915)	(0.957)
Seasonal physical shock (placebo)	-0.052	-0.097	+0.012	-0.093	-0.129	-0.036
	(0.827)	(0.636)	(0.959)	(0.598)	(0.435)	(0.808)
Hist. disaster frequency (host, dynamic)	-0.020	-0.072	-0.053	+0.019	+0.030	+0.049
	(0.749)	(0.226)	(0.389)	(0.752)	(0.596)	(0.357)
Observations (Log-OLS)	*11,871*	*11,501*	*11,279*	*11,119*	*10,919*	*10,652*
Panel B: PPML Estimates						
Sudden physical shock	-0.056	-0.375	-0.148	-0.524	-0.389	-0.765**
	(0.876)	(0.235)	(0.707)	(0.171)	(0.399)	(0.042)
Sudden shock × Hist. disaster freq.	+0.485	-0.015	-0.964	+1.151	+1.438	+0.976
	(0.664)	(0.990)	(0.543)	(0.334)	(0.288)	(0.416)
Seasonal physical shock (placebo)	-0.326	-0.432*	+0.009	-0.026	+0.115	-0.049
	(0.111)	(0.050)	(0.983)	(0.941)	(0.653)	(0.762)
Hist. disaster frequency (host, dynamic)	-0.044	-0.082	-0.072	-0.060	+0.022	-0.007
	(0.509)	(0.155)	(0.255)	(0.387)	(0.715)	(0.901)
Observations (PPML)	*13,210*	*13,444*	*13,501*	*13,530*	*13,512*	*13,287*
Model Specification & Statistics						
Macroeconomic & Governance Controls	Yes	Yes	Yes	Yes	Yes	Yes
Dyad Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad	Dyad	Dyad

Appendix F RQ3: Robustness checks

Appendix - Table xi: Robustness Check I: ENT OLS

Variables	T=0	T+1	T+2	T+3
Disaster Shocks & GVC Interactions				
Disaster severity shock (D_t)	+0.308***	+0.308***	+0.126	+0.033
	(0.006)	(0.009)	(0.266)	(0.803)
GVC position index ($G_{i,s}$, centered)	+0.081	+0.052	+0.105	+0.226
	(0.478)	(0.667)	(0.400)	(0.106)
Hist. disaster exposure ($H_{i,t}$, centered)	+0.013	+0.013	+0.066	+0.056
	(0.769)	(0.777)	(0.145)	(0.243)
Disaster severity × GVC position ($D \times G$)	-1.085*	-1.595***	-0.407	-0.448
	(0.068)	(0.006)	(0.453)	(0.598)
Disaster severity × Hist. exposure ($D \times H$)	-0.474	-0.921*	-0.764	-0.679*
	(0.339)	(0.066)	(0.154)	(0.219)
GVC position × Hist. exposure ($G \times H$)	+0.207	+0.268*	+0.239	+0.185
	(0.164)	(0.092)	(0.154)	(0.320)
Disaster × GVC × Hist. exposure ($D \times G \times H$)	**-0.457**	**+0.822**	**+2.956**	**+4.955**
	(0.872)	(0.775)	(0.355)	(0.140)
Macroeconomic & Governance Controls				
Log GDP per capita (host, $t-1$)	+0.237***	+0.216***	+0.098	-0.035
	(0.001)	(0.003)	(0.198)	(0.667)
Log GDP per capita (home, $t-1$)	+0.431***	+0.441***	+0.448***	+0.274**
	(0.000)	(0.000)	(0.000)	(0.041)
Gov. effectiveness (host, $t-1$, centered)	-0.063**	-0.064**	-0.045	-0.028
	(0.018)	(0.020)	(0.115)	(0.348)
Model Specification & Statistics				
Dyad Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad
Observations	50,667	42,324	36,298	30,960
Dyad-years with disaster exposure	32,915	27,326	23,312	19,574
Distinct disaster country-years	409	372	337	299

*Notes: Log-OLS estimates across Local Projections horizons $T = 0$ to $T + 3$ using Strategy B (NACE section level merged with OECD TiVA data). p-values in parentheses derived from standard errors clustered at the dyad level. *** $p < 0.01$, $p < 0.05$, * $p < 0.1$. Continuous variables participating in interactions (G, H, Gov. Eff.) are mean-centered*

Appendix - Table xii: Robustness II: PPML EMP

Variables	T=0	T+1	T+2	T+3
Disaster Shocks & GVC Interactions				
Disaster severity shock (D_t)	-0.544*	-0.213	-0.265	+0.019
	(0.051)	(0.206)	(0.141)	(0.904)
GVC position index ($G_{i,s}$, centered)	-0.490	-0.266	+0.137	+0.282
	(0.102)	(0.334)	(0.547)	(0.224)
Hist. disaster exposure ($H_{i,t}$, centered)	+0.351***	+0.295**	+0.333***	+0.292***
	(0.006)	(0.014)	(0.001)	(0.001)
Disaster severity × GVC position ($D \times G$)	-0.728	+0.206	-0.584	+0.455
	(0.553)	(0.801)	(0.457)	(0.506)
Disaster severity × Hist. exposure ($D \times H$)	-1.341	-1.544	-0.690	-0.232
	(0.211)	(0.102)	(0.452)	(0.739)
GVC position × Hist. exposure ($G \times H$)	+0.753*	+0.259	+0.219	+0.118
	(0.051)	(0.440)	(0.476)	(0.732)
Disaster × GVC × Hist. exposure ($D \times G \times H$)	**-4.341**	**+4.506**	**+1.848**	**+2.225**
	(0.586)	(0.548)	(0.783)	(0.685)
Macroeconomic & Governance Controls				
Log GDP per capita (host, $t-1$)	+0.605***	+0.602***	+0.318***	+0.078
	(0.000)	(0.000)	(0.007)	(0.518)
Log GDP per capita (home, $t-1$)	+0.650**	+0.529**	+0.726***	+0.761***
	(0.034)	(0.050)	(0.004)	(0.001)
Gov. effectiveness (host, $t-1$, centered)	-0.008	-0.014	-0.006	+0.066
	(0.866)	(0.738)	(0.886)	(0.138)
Model Specification & Statistics				
Dyad Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Clustered Standard Errors	Dyad	Dyad	Dyad	Dyad
Observations	44,978	39,889	34,998	30,228
Dyad-years with disaster exposure	29,010	25,407	22,095	18,778
Distinct disaster country-years	409	372	337	299

*Notes: PPML estimates across Local Projections horizons $T = 0$ to $T + 3$ using Strategy B (NACE section level merged with OECD TiVA data). p-values in parentheses derived from standard errors clustered at the dyad level. *** $p < 0.01$, $p < 0.05$, * $p < 0.1$. Continuous variables participating in interactions (G, H, Gov. Eff.) are mean-centered.*

Appendix - Table xiii: Robustness IV: 2008-2016 cohort PPML EMP RQ3

	PPML EMP T=0	PPML EMP T+1	PPML EMP T+2	PPML EMP T+3
Variable				
D (total severity) coef	-0.532	-0.242	-0.294	0.019
	(0.115)	(0.221)	(0.115)	(0.904)
G (GVC position) coef	-0.402	-0.239	0.232	0.282
	(0.133)	(0.367)	(0.319)	(0.224)
H (hist. exposure) coef	+0.395***	+0.318**	+0.355***	+0.292***
	(0.009)	(0.020)	(0.001)	(0.001)
D x G coef	-0.567	0.712	-0.27	0.455
	(0.713)	(0.457)	(0.745)	(0.506)
D x H coef	-0.262	-0.364	-0.108	-0.232
	(0.808)	(0.662)	(0.903)	(0.739)
G x H coef	+0.842*	0.269	0.132	0.118
	(0.059)	(0.450)	(0.681)	(0.732)
D x G x H coef	-3.418	3.474	0.191	2.225
	(0.667)	(0.597)	(0.976)	(0.685)
ln(GDP/cap) host coef	+0.562***	+0.560***	+0.270**	0.078
	(0.004)	(0.000)	(0.036)	(0.518)
ln(GDP/cap) home coef	+0.566*	+0.493*	+0.725***	+0.761***
	(0.076)	(0.071)	(0.004)	(0.001)
Gov. Eff. host coef	0.021	0.012	0.019	0.066
	(0.664)	(0.767)	(0.685)	(0.138)
N (sample)	32,803	32,248	31,491	30,228
Dyad-years w/ disaster	20,482	20,056	19,563	18,778
Distinct disaster ctry-years	299	299	299	299

Appendix - Table xiv: Robustness III: Split-Sample Heterogeneity Analysis by Historical Disaster Exposure - Dependent Variable: Sectoral Affiliate Activity (ENT)

Group	Horizon	OLS N	OLS β_D	OLS β_G	OLS β_DxG	PPML N	PPML β_D	PPML β_G	PPML β_DxG
Low exposure (Below Median)	T=0	26.099	+0.404*	-0.217	-1.454	33.205	+0.215	-0.183	-2.704**
			(0.050)	(0.174)	(0.212)		(0.437)	(0.399)	(0.036)
	T+1	21.875	+0.407**	-0.250	-2.056*	29.667	+0.234	-0.103	-4.281***
			(0.050)	(0.148)	(0.060)		(0.365)	(0.622)	(0.000)
	T+2	18.765	+0.065	-0.126	-2.067	26.179	+0.094	-0.026	-3.311**
			(0.783)	(0.492)	(0.140)		(0.720)	(0.900)	(0.014)
	T+3	16.03	+0.108	+0.047	-2.960**	22.761	+0.373	+0.118	-1.541
			(0.633)	(0.812)	(0.024)		(0.244)	(0.589)	(0.368)
High exposure (Above Median)	T=0	24.568	+0.295*	+0.377**	-1.130	31.468	+0.128	+0.529	+0.047
			(0.053)	(0.019)	(0.128)		(0.355)	(0.146)	(0.948)
	T+1	20.449	+0.260*	+0.355**	-1.573**	28.087	+0.059	+0.401	+0.700
			(0.095)	(0.032)	(0.029)		(0.718)	(0.319)	(0.350)
	T+2	17.533	+0.102	+0.343**	-0.100	24.783	-0.162	+0.881***	+0.329
			(0.464)	(0.045)	(0.874)		(0.432)	(0.004)	(0.695)
	T+3	14.93	-0.085	+0.401**	+0.351	21.516	-0.185	+0.953***	+0.470
			(0.637)	(0.042)	(0.751)		(0.147)	(0.007)	(0.509)

*Notes: Subsamples are split at the sample median of host country historical disaster exposure ($H_{i,t}$). All regressions include macro controls (ln GDP $pc_{host,\,t-1}$, ln GDP $pc_{home,\,t-1}$, Gov. $Eff._{host,\,t-1}$) and dyad + year fixed effects. p-values in parentheses derived from standard errors clustered at the dyad level. *** $p < 0.01$, $p < 0.05$, * $p < 0.1$. Continuous variables are mean-centered.*

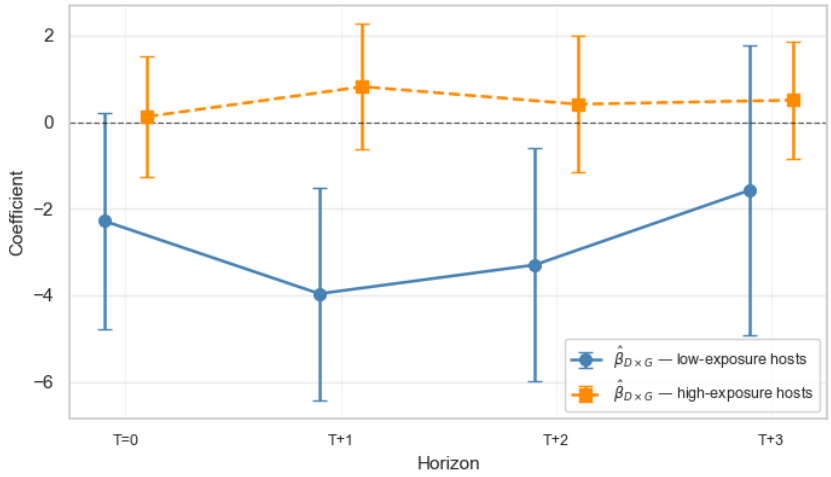


Figure 0-1: Disaster × value-chain position interaction, estimated separately for low- and high-exposure hosts

To probe the sensitivity of the main triple-interaction estimate to its continuous specification, the sample is split into two subgroups of host countries — "high-exposure" and "low-exposure" — based on the median of host-level mean historical disaster severity, and the pooled model is re-estimated with discrete group-interaction dummies (D × High, G × High, D×G × High) replacing the continuous moderator H. The motivation is statistical power: the continuous triple interaction D×G×H asks the data to trace out a smooth interaction surface across a moderator that varies only across the 56 TiVA-eligible host countries, so its precision is sensitive to a small number of extreme-H observations. A discrete split concentrates the test on the sharper contrast between two exposure regimes and additionally offers a directly readable set of group-specific coefficients, at the cost of discarding within-group variation in H.

The split reveals a pattern that departs from what the continuous specification implied. In the PPML estimation, the D×G interaction is large, negative, and significant among low-exposure hosts at T=0 (−2.704, p=0.036), T+1 (−4.281, p<0.001) and T+2 (−3.311, p=0.014), fading to insignificance by T+3 (−1.541, p=0.368); the same interaction is small and never significant among high-exposure hosts (from +0.047 to +0.700, all p>0.35). A second, unplanned pattern also emerges on the GVC main effect taken alone: β_G turns significant and positive specifically in the high-exposure group at T+2 (+0.881, p=0.004) and T+3 (+0.953, p=0.007), while remaining statistically null at T=0/T+1 in that same group and throughout the low-exposure group at every horizon. Neither of these patterns was part of the pre-specified RQ3 estimand, which targets the triple interaction rather than either constituent term in isolation.

Taken as a robustness check on the main test, this split does not straightforwardly corroborate it: the main continuous estimate in Section 4.4.1 was statistically significant at one horizon only, yet the discrete alternative surfaces significance mainly through the subgroup-specific D×G term rather than through the formal high-versus-low difference term that is its direct discrete analogue — the two specifications diverge rather than converge. The β_G pattern is a further reason for caution: it is significant only at T+2/T+3 with no a priori theoretical reason to expect this particular timing, and it does not survive replication under OLS, which instead finds β_G significant at every horizon in the high-exposure group (+0.343 to +0.401, all p<0.05) — an estimator-dependent result is more consistent with sampling variability than with a genuine mechanism. This is compounded by the small number of clusters underlying inference in each subgroup: with roughly 28 host countries per group, dyad-clustered standard errors rest on a level of identifying variation well below what asymptotic cluster-robust theory is generally considered reliable for. Verifying whether either pattern survives host-level clustering, a wild cluster bootstrap suited to few clusters (Cameron, C., & Miller, D., 2015), randomization inference, and an explicit multiple-comparison correction across the eight subgroup-horizon cells would be the natural next steps; these checks are, however, left outside the scope of the present thesis.